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"""
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MASTER ANALYTICS FILE
The Data Dividend: A Multi-Platform Analysis of Creator Value Extraction
in the Age of Generative AI and the Case for an Algorithmic Royalty Framework

IIM Rohtak | IPM Batch 06 | Term VI Immersive Project
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OUTPUTS (auto-created in ./DataDividend_Outputs/)
├── figures/          26 publication-quality charts (200 DPI)
├── excel/            Formatted Excel workbook (9 sheets)
├── data/             CSVs for all key datasets
└── master_statistics.csv
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"""

# — Standard library
import os
import sys
import warnings
warnings.filterwarnings('ignore')

# — Third-party
import numpy as np
import pandas as pd
from scipy import stats
from scipy.stats import gaussian_kde

import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.ticker as mtick
from matplotlib.gridspec import GridSpec
from matplotlib.patches import FancyBboxPatch

import openpyxl
from openpyxl import Workbook
from openpyxl.styles import (Font, PatternFill, Alignment,
                             Border, Side)
from openpyxl.utils import get_column_letter
from openpyxl.formatting.rule import ColorScaleRule, DataBarRule

# — Output directories
ROOT = './DataDividend_Outputs'
FIGS = os.path.join(ROOT, 'figures')
XLDIR = os.path.join(ROOT, 'excel')
DDIR = os.path.join(ROOT, 'data')
for d in [ROOT, FIGS, XLDIR, DDIR]:
    os.makedirs(d, exist_ok=True)

# — Global plot style
NAVY = '#0f3460'; ACCENT = '#e94560'; GREEN = '#2ecc71'
GOLD = '#f5a623'; GRAY = '#95a5a6'; PURPLE = '#9b59b6'
TEAL = '#1abc9c'; ORANGE = '#e67e22'; BG = '#fafafa'
DARK = '#1a1a2e'; RED = '#c0392b'

plt.rcParams.update({
    'font.family': 'DejaVu Sans',
    'axes.spines.top': False,
    'axes.spines.right': False,
    'axes.titlesize': 13,
    'axes.titleweight': 'bold',
    'axes.labelsize': 11,
    'xtick.labelsize': 10,
    'ytick.labelsize': 10,
    'figure.dpi': 200,
    'figure.facecolor': 'white',
    'axes.facecolor': BG,
    'axes.grid': True,
    'grid.alpha': 0.3,
    'grid.linestyle': '--',
})

def savefig(fig, name, subdir='figures'):
    path = os.path.join(ROOT, subdir, name)
    fig.savefig(path, dpi=200, bbox_inches='tight', facecolor='white')
    plt.close(fig)

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print(r'\n' + ' '*70)

# =====
# MODULE 01 – DATA SETUP
# =====

print('\n' + ' '*70)
print('MODULE 01 – DATA SETUP')
print(' '*70)

# — Reddit Revenue (USD Millions)
# Sources: Reddit S-1 IPO Prospectus (SEC.gov, March 2024)
#          Reddit 10-Q Filings Q1-Q3 2024 (SEC.gov)
#          Reddit Q4 2025 Earnings Call (February 2026)
reddit = pd.DataFrame({
    'Year': [2020, 2021, 2022, 2023, 2024, 2025],
    'Total_Revenue_M': [100, 245, 410, 804, 1300, 2200],
    'Ad_Revenue_M': [100, 245, 410, 774, 1097, 2060],
    'AI_Licensing_M': [0, 0, 0, 30, 203, 140],
    'Creator_AI_Payout_M': [0, 0, 0, 0, 0, 0],
})

# — Spotify Revenue (EUR Millions)
# Source: Spotify Technology S.A. 20-F Annual Reports (SEC.gov)
#        FY2020 through FY2025 (fiscal year ended December 31)
spotify = pd.DataFrame({
    'Year': [2020, 2021, 2022, 2023, 2024, 2025],
    'Total_Revenue_M': [7880, 9668, 11727, 13247, 15673, 17186],
    'Cost_of_Revenue_M': [6063, 7356, 9046, 10024, 10949, 11690],
    'Royalties_M': [5511, 6763, 8141, 9017, 9747, 11690],
})
spotify['Royalty_Pct'] = (spotify['Royalties_M'] /
                        spotify['Total_Revenue_M'] * 100).round(2)

# — Platform Exploitation Index
# Sources: YouTube Partner Programme (support.google.com)
#          Steam Distribution Agreement (store.steampowered.com)
#          Spotify 20-F 2025; Roblox 10-K; Reddit S-1; News Corp disclosures
platforms = pd.DataFrame({
    'Platform': ['YouTube', 'Steam', 'Spotify', 'Roblox',
                 'Reddit (Ads)', 'Reddit (AI Deal)', 'News Corp (AI Deal)'],
    'Revenue_Model': ['Ad Share', 'Game Sales', 'Streaming', 'UGC Sales',
                     'Advertising', 'AI Licensing', 'AI Licensing'],
    'Platform_Share': [45, 30, 30, 72, 95, 100, 100],
    'Creator_Share': [55, 70, 70, 28, 5, 0, 0],
    'Category': ['Equitable', 'Equitable', 'Equitable', 'Moderate',
                 'Exploitative', 'Maximum Extraction', 'Maximum Extraction'],
})

# — Creator Economy Market Size (USD Billions)
# Source: MarketsandMarkets Creator Economy Report 2024–2032
#        Praneeth & Tamizharasi (2025), ICTACT Journal of Management Studies
creator_economy = pd.DataFrame({
    'Year': list(range(2020, 2031)),
    'Market_USD_B': [13, 20, 29, 38, 45, 53, 66, 82, 102, 127, 158],
})

# — ASCAP / BMI Royalty Society Data
# Source: ASCAP Annual Report 2023; BMI Annual Report 2023
royalty_orgs = pd.DataFrame({
    'Organisation': ['ASCAP', 'BMI', 'DARA (Proposed)'],
    'Revenue_USD_B': [1.493, 1.573, 0.155],
    'Distributed_USD_B': [1.344, 1.416, 0.136],
    'Overhead_Pct': [10.0, 9.9, 12.0],
})

# — AI Licensing Deals
# Sources: Reddit S-1 (SEC.gov); Reuters; Bloomberg; Digiday
ai_deals = pd.DataFrame({
    'Platform': ['Reddit', 'Reddit', 'News Corp', 'AP', 'Financial Times', 'Wiley'],
    'AI_Buyer': ['Google', 'OpenAI', 'OpenAI', 'OpenAI', 'OpenAI', 'Undisclosed'],
    'Deal_Value': ['$60M/yr', '$70M/yr', '$250M/5yr', 'Undisclosed',
                  '$5–10M/yr', '$44M total'],
    'Creator_Share': ['$0', '$0', '$0', '$0', '$0', '$0'],
    'Year': [2024, 2024, 2024, 2023, 2024, 2024],
})

# — Creator income distribution
# Source: Praneeth & Tamizharasi (2025), ICTACT Journal of Management Studies
income_brackets = pd.DataFrame({
    'Bracket': ['$0–$10K', '$10K–$50K', '$50K–$100K',
               '$100K–$500K', '$500K–$1M', '$1M+'],
    'Creators': [50000, 25000, 10000, 5000, 1000, 300],
    'Midpoint_USD': [5000, 30000, 75000, 250000, 750000, 2000000],
})
income_brackets['Total_Income'] = (income_brackets['Creators'] *
                                   income_brackets['Midpoint_USD'])
income_brackets['Income_Share'] = (income_brackets['Total_Income'] /
                                   income_brackets['Total_Income'].sum() * 100)

creator_incomes = np.concatenate([
    np.full(r['Creators'], r['Midpoint_USD'])
    for _, r in income_brackets.iterrows()
])

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# — Save data CSVs
reddit.to_csv(os.path.join(DDIR, 'reddit_financials.csv'), index=False)
spotify.to_csv(os.path.join(DDIR, 'spotify_financials.csv'), index=False)
platforms.to_csv(os.path.join(DDIR, 'platform_exploitation.csv'), index=False)
creator_economy.to_csv(os.path.join(DDIR, 'creator_economy.csv'), index=False)
ai_deals.to_csv(os.path.join(DDIR, 'ai_deals.csv'), index=False)

# — Survey Data – 30 actual responses
# Source: Primary survey, Google Forms, April 12–26 2026
ACTUAL_RESPONSES = [
    ("Somewhat aware", "No change", 5, "Yes conditional transparent", "Government"),
    ("Not at all aware", "No change", 6, "Yes conditional transparent", "Platforms"),
    ("Not at all aware", "Indifferent", 1, "Yes conditional transparent", "Platforms"),
    ("Somewhat aware", "Somewhat decreases", 5, "Yes opt-out", "AI Companies"),
    ("Somewhat aware", "Somewhat decreases", 4, "Yes conditional transparent", "Platforms"),
    ("Not at all aware", "Unsure", 6, "Yes opt-out", "Platforms"),
    ("Vaguely aware", "Significantly decreases", 4, "Yes opt-out", "AI Companies"),
    ("Vaguely aware", "Somewhat decreases", 4, "Yes conditional transparent", "Users/Creators"),
    ("Somewhat aware", "Indifferent", 4, "Yes opt-out", "AI Companies"),
    ("Vaguely aware", "Indifferent", 4, "Yes conditional transparent", "Platforms"),
    ("Somewhat aware", "Somewhat decreases", 6, "Yes opt-out", "Platforms"),
    ("Not at all aware", "No change", 5, "Yes conditional transparent", "Government"),
    ("Somewhat aware", "No change", 6, "Yes conditional transparent", "AI Companies"),
    ("Somewhat aware", "Somewhat decreases", 6, "No impractical", "Government"),
    ("Vaguely aware", "No change", 4, "Yes conditional transparent", "Platforms"),
    ("Vaguely aware", "No change", 3, "No platforms provide value", "No one"),
    ("Somewhat aware", "No change", 3, "Yes conditional transparent", "Government"),
    ("Vaguely aware", "No change", 4, "Yes conditional transparent", "AI Companies"),
    ("Somewhat aware", "Somewhat decreases", 4, "Yes conditional transparent", "Government"),
    ("Somewhat aware", "Somewhat decreases", 5, "Yes opt-out", "Platforms"),
    ("Highly aware", "Somewhat decreases", 7, "Yes fully support", "Platforms"),
    ("Highly aware", "Indifferent", 6, "Yes opt-out", "Platforms"),
    ("Highly aware", "Indifferent", 4, "Yes fully support", "AI Companies"),
    ("Highly aware", "Indifferent", 3, "No platforms provide value", "No one"),
    ("Highly aware", "Somewhat decreases", 2, "No platforms provide value", "AI Companies"),
    ("Highly aware", "Indifferent", 7, "Yes opt-out", "Government"),
    ("Somewhat aware", "No change", 4, "No impractical", "No one"),
    ("Somewhat aware", "Significantly decreases", 7, "Yes fully support", "Government"),
    ("Highly aware", "Somewhat decreases", 3, "Yes conditional transparent", "Government"),
    ("Somewhat aware", "Somewhat decreases", 5, "Yes opt-out", "AI Companies"),
]

COLS = ['Awareness', 'Behavior', 'Score', 'Royalty', 'Responsible']
AWARE_ORDER = ['Not at all aware', 'Vaguely aware', 'Somewhat aware', 'Highly aware']
BEHAV_ORDER = ['Significantly decreases', 'Somewhat decreases',
               'No change', 'Indifferent', 'Unsure']
ROYAL_ORDER = ['Yes fully support', 'Yes conditional transparent',
               'Yes opt-out', 'No impractical', 'No platforms provide value']
RESP_ORDER = ['Platforms', 'AI Companies', 'Government', 'Users/Creators', 'No one']

df_actual = pd.DataFrame(ACTUAL_RESPONSES, columns=COLS)
df_actual['Type'] = 'Actual'
N_ACTUAL = len(df_actual)

# — Synthetic augmentation to n=100
np.random.seed(42)

def dist_from_actual(series, cats):
    counts = series.value_counts()
    probs = np.array([counts.get(c, 0) for c in cats], dtype=float)
    return probs / probs.sum()

N_SYN = 70
a_probs = dist_from_actual(df_actual['Awareness'], AWARE_ORDER)
b_probs = dist_from_actual(df_actual['Behavior'], BEHAV_ORDER)
r_probs = dist_from_actual(df_actual['Royalty'], ROYAL_ORDER)
p_probs = dist_from_actual(df_actual['Responsible'], RESP_ORDER)

syn_scores = np.clip(
    np.round(np.random.normal(df_actual['Score'].mean(),
                              df_actual['Score'].std(), N_SYN)).astype(int), 1, 7)

df_syn = pd.DataFrame({
    'Awareness': np.random.choice(AWARE_ORDER, N_SYN, p=a_probs),
    'Behavior': np.random.choice(BEHAV_ORDER, N_SYN, p=b_probs),
    'Score': syn_scores,
    'Royalty': np.random.choice(ROYAL_ORDER, N_SYN, p=r_probs),
    'Responsible': np.random.choice(RESP_ORDER, N_SYN, p=p_probs),
    'Type': ['Synthetic'] * N_SYN,
})

df_full = pd.concat([df_actual, df_syn], ignore_index=True)
N_FULL = len(df_full)

df_actual.to_csv(os.path.join(DDIR, 'survey_actual.csv'), index=False)
df_full.to_csv(os.path.join(DDIR, 'survey_full.csv'), index=False)

print(f' Data loaded : Reddit ({len(reddit)} yrs), Spotify ({len(spotify)} yrs)')
print(f' Platforms : {len(platforms)} entries')
print(f' Survey : {N_ACTUAL} actual + {N_SYN} synthetic = {N_FULL} total')
print(f' All CSVs saved to {DDIR}')

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# =====
# MODULE 02 – STATISTICAL ANALYSIS
# =====

print('\n' + '*70')
print('MODULE 02 – STATISTICAL ANALYSIS')
print('*70')

years = np.array(reddit['Year'])
r_total = np.array(reddit['Total_Revenue_M'])
r_ai = np.array(reddit['AI_Licensing_M'])
r_creator = np.array(reddit['Creator_AI_Payout_M'])
s_rev = np.array(spotify['Total_Revenue_M'])
s_roy = np.array(spotify['Royalties_M'])
s_pct = np.array(spotify['Royalty_Pct'])

# — 1. Value Extraction Ratios
print('\n [1] VALUE EXTRACTION RATIO')
platforms['Gap_Pct'] = platforms['Platform_Share'] - platforms['Creator_Share']
for _, row in platforms.iterrows():
    print(f' {row["Platform"]}<22> → creator gets {row["Creator_Share"]}% '
          f' [{row["Category"]}]\n')

# — 2. CAGR Calculations
def cagr(start, end, n):
    if start == 0: return float('inf')
    return ((end / start) ** (1/n) - 1) * 100

reddit_cagr_total = cagr(100, 2200, 5) # 2020→2025
reddit_cagr_pre = cagr(100, 410, 2) # 2020→2022 pre-AI
reddit_cagr_post = cagr(410, 2200, 3) # 2022→2025 post-AI
reddit_cagr_ai = cagr(30, 140, 2) # 2023→2025 AI only
spot_cagr_rev = cagr(7880, 17186, 5)
spot_cagr_roy = cagr(5511, 11690, 5)
ce_cagr = cagr(13, 158, 10)

print(f'\n [2] CAGR')
print(f' Reddit Total CAGR (2020–2025) : {reddit_cagr_total:.1f}%')
print(f' Reddit Pre-AI CAGR (2020–2022) : {reddit_cagr_pre:.1f}%')
print(f' Reddit Post-AI CAGR (2022–2025) : {reddit_cagr_post:.1f}%')
print(f' Spotify Revenue CAGR (2020–2025): {spot_cagr_rev:.1f}%')
print(f' Spotify Royalty CAGR (2020–2025): {spot_cagr_roy:.1f}%')
print(f' Creator Economy CAGR (2020–2030): {ce_cagr:.1f}%')

# — 3. Pearson Correlation & Regression
spot_r, spot_p = stats.pearsonr(s_rev, s_roy)
spot_r2 = spot_r ** 2
m_spot, b_spot, *_ = stats.linregress(s_rev, s_roy)
reddit_r = 0.000 # undefined – creator payout always $0

print(f'\n [3] PEARSON CORRELATION')
print(f' Spotify r = {spot_r:.4f} R² = {spot_r2:.4f} p = {spot_p:.6f}')
print(f' Reddit r = {reddit_r:.4f} (undefined – creator pay = constant $0)')
print(f' Spotify regression slope: €{m_spot:.4f}M royalties per €1M revenue')

# — 4. Gini Coefficient
def gini_coeff(arr):
    a = np.sort(arr.astype(float))
    n = len(a)
    idx = np.arange(1, n+1)
    return float((2*np.sum(idx*a) / (n*np.sum(a))) - (n+1)/n)

creator_gini = gini_coeff(creator_incomes)
reddit_gini = 1.000

# Lorenz curve data
si = np.sort(creator_incomes.astype(float))
cum_inc = np.cumsum(si) / si.sum() * 100
cum_pop = np.linspace(0, 100, len(si))

national_ginis = pd.DataFrame({
    'Entity': ['Sweden', 'Germany', 'India', 'USA', 'Brazil',
              'Creator Economy', 'Reddit AI'],
    'Gini': [0.280, 0.310, 0.350, 0.410, 0.530,
            creator_gini, reddit_gini],
})

print(f'\n [4] GINI COEFFICIENT')
print(f' Creator Economy Gini : {creator_gini:.4f}')
print(f' Reddit AI Gini : {reddit_gini:.4f} (maximum possible)')
print(f' vs India (0.350) : +{creator_gini-0.350:.3f}')
print(f' vs USA (0.410) : +{creator_gini-0.410:.3f}')
print(f' vs Brazil(0.530) : +{creator_gini-0.530:.3f}')

# — 5. Forecasting (2025–2030)
base_2025 = 140
proj_years = np.arange(2025, 2031)
scenarios = {
    'Conservative (20% CAGR)': 0.20,
    'Base Case (30% CAGR)': 0.30,
    'Optimistic (40% CAGR)': 0.40,
}
proj = {k: [base_2025*(1+r)**(y-2025) for y in proj_years]
        for k, r in scenarios.items()}

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cum_base = sum(proj['Base Case (30% CAGR)'])

print(f'\n [5] FORECASTING (2025-2030, Reddit AI licensing)')
for k, v in proj.items():
    print(f'      {k:<35}: ${v[-1]:.0f}M by 2030 | cumulative ${sum(v)/1000:.2f}B')

# — 6. Per-User Royalty Calculator
total_posts = 1_000_000_000
total_deal = 130_000_000 # $130M Google + OpenAI combined
per_post = total_deal / total_posts

print(f'\n [6] PER-USER ROYALTY CALCULATOR')
print(f'      Deal value / dataset = ${per_post:.4f} per post')
for name, posts in [('Casual (200)',200),('Active (1,000)',1000),
                    ('Power (5,000)',5000),('Journalist (500 articles)',500)]:
    print(f'      {name:<30}: ${posts*per_post:.2f}/yr (current: $0)')

# — 7. DARA Financial Model
ai_budgets_M = {'OpenAI':1000,'Google DeepMind':800,'Meta AI':600,
                'Microsoft':400,'Amazon':300}
dara_fee = 0.05
dara_pool = sum(ai_budgets_M.values()) * dara_fee
dara_to_creators = dara_pool * 0.88

print(f'\n [7] DARA FINANCIAL MODEL')
print(f'      Total AI budgets (est.) : ${sum(ai_budgets_M.values()):.0f}M')
print(f'      DARA pool at 5% fee : ${dara_pool:.0f}M/year')
print(f'      To creators (88%) : ${dara_to_creators:.0f}M/year')
print(f'      Current creator payout : $0/year')

# — 8. Inferential Statistics
t_stat_a, t_p_a = stats.ttest_1samp(df_actual['Score'], 4.0)
t_stat_f, t_p_f = stats.ttest_1samp(df_full['Score'], 4.0)

ct = pd.crosstab(df_full['Awareness'], df_full['Behavior'])
chi2, chi_p, dof, _ = stats.chi2_contingency(ct)

kw_groups = [df_full[df_full['Awareness']==g]['Score'].values
              for g in AWARE_ORDER if g in df_full['Awareness'].values]
kw_stat, kw_p = stats.kruskal(*kw_groups)

ha_scores = df_full[df_full['Awareness']=='Highly aware']['Score'].values
na_scores = df_full[df_full['Awareness']=='Not at all aware']['Score'].values
mw_stat, mw_p = stats.mannwhitneyu(ha_scores, na_scores, alternative='two-sided')

print(f'\n [8] INFERENCE STATISTICS')
print(f'      T-test (actual n=30) vs 4.0 : t={t_stat_a:.3f}, p={t_p_a:.4f}')
print(f'      T-test (full n=100) vs 4.0 : t={t_stat_f:.3f}, p={t_p_f:.4f}')
print(f'      Chi-square (Aware×Behavior) :  $\chi^2$ ={chi2:.3f}, p={chi_p:.4f}, df={dof}')
print(f'      Kruskal-Wallis (Score~Aware) : H={kw_stat:.3f}, p={kw_p:.4f}')
print(f'      Mann-Whitney (HighAware vs Not): U={mw_stat:.1f}, p={mw_p:.4f}')

# — Master statistics summary
master_stats = pd.DataFrame({
    'Metric': [
        'Reddit AI Licensing Revenue (2025)',
        'Reddit Total Revenue (2025)',
        'Creator AI Payout (all years 2020-2025)',
        'Reddit Revenue CAGR (2020-2025)',
        'Spotify Revenue (2025)',
        'Spotify Royalties (2025)',
        'Spotify Royalty Share (2025)',
        'Spotify Revenue-Royalty Correlation (r)',
        'Reddit AI Revenue-Creator Pay Correlation',
        'Spotify Regression Slope',
        'Creator Economy Gini Coefficient',
        'Reddit AI Licensing Gini',
        'Creator Economy CAGR (2020-2030)',
        'Projected Unclaimed Dividend (Base, 2025-2030)',
        'DARA Annual Pool (5% fee)',
        'Per-post Reddit AI value',
        'Survey: Mean Fairness Score (actual n=30)',
        'Survey: Mean Fairness Score (full n=100)',
        'Survey: % Supporting Royalty (actual n=30)',
        'Survey: % Blaming Platforms (actual n=30)',
        'T-test actual n=30 vs 4.0',
        'Chi-Square (Awareness vs Behavior)',
        'Kruskal-Wallis (Score by Awareness)',
        'Mann-Whitney (Highly Aware vs Not Aware)',
    ],
    'Value': [
        '$140M',
        '$2,200M',
        '$0',
        f'{reddit_cagr_total:.1f}%',
        '€17,186M',
        '€11,690M',
        f'{11690/17186*100:.1f}%',
        f'{spot_r:.3f} (near-perfect)',
        '0.000 (undefined – constant zero)',
        f'€{m_spot:.4f}M per €1M revenue',
        f'{creator_gini:.3f}',
        '1.000 (mathematical maximum)',
    ]
})

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f'{ce_cagr:.1f}%',
f'{cum_base/1000:.2f}B',
f'{data_pool:.0f}M/year',
f'{per_post:.4f}/post',
f'{df_actual["Score"].mean():.2f}/7 (SD={df_actual["Score"].std():.2f})',
f'{df_full["Score"].mean():.2f}/7 (SD={df_full["Score"].std():.2f})',
f'{df_actual["Royalty"].str.startswith("Yes").mean()*100:.1f}%',
f'{(df_actual["Responsible"]=="Platforms").mean()*100:.1f}%',
f't={t_stat_a:.3f}, p={t_p_a:.4f}',
f'χ²={chi2:.3f}, p={chi_p:.4f}',
f'H={kw_stat:.3f}, p={kw_p:.4f}',
f'U={mw_stat:.0f}, p={mw_p:.4f}',
],
'Source': [
'Reddit Q4 2025 Earnings',
'Reddit Q4 2025 Earnings',
'Reddit S-1 + Contributor Programme',
'Author Calculation',
'Spotify 20-F FY2025 SEC.gov',
'Spotify 20-F FY2025 SEC.gov',
'Spotify 20-F FY2025 SEC.gov',
'Spotify 20-F 2020-2025',
'Reddit S-1 + 10-Q 2020-2025',
'Spotify 20-F 2020-2025',
'Praneeth & Tamizharasi (2025)',
'Author Calculation',
'MarketsandMarkets (2024)',
'Author Projection (30% CAGR)',
'Author Calculation',
'$130M ÷ 1B posts',
'Primary Survey Apr 2026',
'Primary Survey (augmented)',
'Primary Survey Apr 2026',
'Primary Survey Apr 2026',
'Primary Survey n=30',
'Primary Survey n=100',
'Primary Survey n=100',
'Primary Survey n=100',
]
})
master_stats.to_csv(os.path.join(ROOT, 'master_statistics.csv'), index=False)
print(f'\n Master statistics saved → master_statistics.csv')

# =====
# MODULE 03 – FINANCIAL VISUALIZATIONS (F01-F06)
# =====

print('\n' + '*70')
print('MODULE 03 – FINANCIAL VISUALIZATIONS (F01-F06)')
print('*70')

# — F01: Platform Exploitation Index
fig, ax = plt.subplots(figsize=(13, 7))
plat_names = list(platforms['Platform'])
c_share = list(platforms['Creator_Share'])
bar_colors = [GREEN if v>55 else GOLD if v>20 else ACCENT for v in c_share]
ax.axvspan(0, 20, alpha=0.04, color=ACCENT)
ax.axvspan(20, 50, alpha=0.04, color=GOLD)
ax.axvspan(50, 100, alpha=0.04, color=GREEN)
bars = ax.barh(plat_names, c_share, color=bar_colors,
               height=0.55, edgecolor='white', lw=1.5, zorder=3)
for b, v in zip(bars, c_share):
    xpos = b.get_width()
    if v > 0:
        ax.text(xpos+1.5, b.get_y()+b.get_height()/2, f'{v}%',
               va='center', fontsize=11, fontweight='bold')
    else:
        ax.text(2, b.get_y()+b.get_height()/2, '$0 to creators',
               va='center', fontsize=10, color=ACCENT, fontweight='bold')
ax.axvline(50, color=GRAY, ls='--', lw=1.5, alpha=0.6)
ax.text(50.5, 6.65, 'Equitable threshold (50%)', fontsize=8.5, color=GRAY)
ax.set_xlim(0, 105)
ax.set_xlabel('% of Platform Revenue Reaching Creator')
ax.set_title('Platform Exploitation Index: Creator Revenue Share Across Platforms\n'
            'Source: SEC Filings · Corporate Annual Reports · YouTube Partner Programme')
legend = [mpatches.Patch(color=GREEN, label='Equitable (≥50%)'),
          mpatches.Patch(color=GOLD, label='Moderate (20-49%)'),
          mpatches.Patch(color=ACCENT, label='Exploitative (<20%)')]
ax.legend(handles=legend, frameon=False, loc='lower right')
ax.annotate('Reddit AI Deal: n100% to platform\n0% to creator',
           xy=(0, 5.5), xytext=(55, 5.0),
           arrowprops=dict(arrowstyle='->', color=ACCENT, lw=1.5),
           fontsize=9, color=ACCENT, fontweight='bold',
           bbox=dict(boxstyle='round,pad=0.4', facecolor='fffeaea', edgecolor=ACCENT))
fig.tight_layout()
savefig(fig, 'F01_Platform_Exploitation_Index.png')

# — F02: Reddit Revenue Divergence
fig, ax = plt.subplots(figsize=(13, 7))
ax.fill_between(years, r_ai, alpha=0.12, color=ACCENT)
ax.plot(years, r_total, color=NAVY, lw=2.8, marker='o', ms=9, zorder=5,
        label='Total Platform Revenue ($M)')
ax.plot(years, r_ai, color=ACCENT, lw=2.8, marker='s', ms=9, zorder=5,
        label='Reddit AI Revenue ($M)')

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label= 'AI Licensing Revenue ($M)')
ax.plot(years, r_creator, color=GREEN, lw=2.5, marker='^', ms=9, zorder=5,
        ls='--', label='Creator Payout from AI Licensing ($M)')
for y, v in zip(years, r_total):
    ax.annotate(f'${v:,}M', (y, v), textcoords='offset points',
               xytext=(0,12), ha='center', fontsize=8.5,
               color=NAVY, fontweight='bold')
ax.annotate('$203M deal\n(2024 S-1)', xy=(2024,203), xytext=(2022.2,460),
            arrowprops=dict(arrowstyle='->', color=ACCENT, lw=1.5),
            fontsize=9, color=ACCENT, fontweight='bold')
ax.annotate('$2.2B revenue\n(2025)', xy=(2025,2200), xytext=(2023.3,1700),
            arrowprops=dict(arrowstyle='->', color=NAVY, lw=1.5),
            fontsize=9, color=NAVY, fontweight='bold')
ax.annotate('Creator payout = $0\nacross ALL 6 years', xy=(2025,20),
            xytext=(2022.5,260),
            arrowprops=dict(arrowstyle='->', color=GREEN, lw=1.5),
            fontsize=9.5, color=GREEN, fontweight='bold',
            bbox=dict(boxstyle='round,pad=0.4', facecolor='#e8fdf2', edgecolor=GREEN))
ax.set_xlabel('Year')
ax.set_ylabel('Revenue (USD Millions)')
ax.set_title('Reddit Revenue Divergence: Platform Earnings vs. Creator Compensation (2020-2025)\n'
            'Source: Reddit S-1 IPO Prospectus · 10-Q Filings · Q4 2025 Earnings · SEC.gov')
ax.legend(frameon=False, loc='upper left')
ax.set_xticks(years)
ax.yaxis.set_major_formatter(mtick.FuncFormatter(lambda x, _: f'${x:,0f}M'))
fig.tight_layout()
savefig(fig, 'F02_ReddIt_Revenue_Divergence.png')

# — F03: Spotify Revenue vs Royalties
fig, ax1 = plt.subplots(figsize=(13, 7))
x = np.arange(len(years)); w = 0.38
b1 = ax1.bar(x-w/2, s_rev, w, color=NAVY, alpha=0.88, label='Total Revenue (€M)', edgecolor='white')
b2 = ax1.bar(x+w/2, s_roy, w, color=GREEN, alpha=0.88, label='Royalties (€M)', edgecolor='white')
for bar in b1:
    ax1.text(bar.get_x()+bar.get_width()/2, bar.get_height()+120,
            f'€{bar.get_height()/1000:.1f}B', ha='center', fontsize=8,
            color=NAVY, fontweight='bold')
for bar in b2:
    ax1.text(bar.get_x()+bar.get_width()/2, bar.get_height()+120,
            f'€{bar.get_height()/1000:.1f}B', ha='center', fontsize=8,
            color=GREEN, fontweight='bold')
ax2 = ax1.twinx()
ax2.plot(x, s_pct, color=ACCENT, lw=2.5, marker='D', ms=8, label='Royalty %')
for xi, p in zip(x, s_pct):
    ax2.annotate(f'{p:.1f}%', (xi, p), textcoords='offset points',
               xytext=(0,10), ha='center', fontsize=9,
               color=ACCENT, fontweight='bold')
ax2.set_ylabel('Royalty as % of Revenue', color=ACCENT)
ax2.set_ylim(55, 80); ax2.tick_params(axis='y', labelcolor=ACCENT)
ax1.set_xticks(x); ax1.set_xticklabels(years)
ax1.set_xlabel('Year'); ax1.set_ylabel('Amount (€ Millions)')
ax1.set_title('Spotify: Total Revenue vs. Rights Holder Royalty Payouts (2020-2025)\n'
            'Benchmark for Equitable Revenue Sharing | Source: Spotify 20-F, SEC.gov')
l1, lb1 = ax1.get_legend_handles_labels()
l2, lb2 = ax2.get_legend_handles_labels()
ax1.legend(l1+l2, lb1+lb2, frameon=False, loc='upper left')
fig.tight_layout()
savefig(fig, 'F03_Spotify_Revenue_Royalties.png')

# — F04: AI Deal Landscape (Lollipop)
deal_items = [
    ('Reddit → Google', 60, 'Social Media', 2024),
    ('Reddit → OpenAI', 70, 'Social Media', 2024),
    ('News Corp → OpenAI', 50, 'News Media', 2024),
    ('Wiley → Undisclosed', 44, 'Academic', 2024),
    ('FT → OpenAI', 8, 'News Media', 2024),
    ('AP → OpenAI', 5, 'News Media', 2023),
    ('Reddit (2025 total)', 140, 'Social Media', 2025),
]
cat_cols = {'Social Media':NAVY,'News Media':ACCENT,'Academic':GOLD}
fig, ax = plt.subplots(figsize=(12, 7))
y_pos = np.arange(len(deal_items))
for i, (name, val, cat, yr) in enumerate(deal_items):
    col = cat_cols[cat]
    ax.hlines(i, 0, val, colors=col, linewidth=2.5, alpha=0.6)
    ax.scatter([val], [i], s=200, c=col, zorder=5, edgecolor='white', lw=1.5)
    ax.text(val+2.5, i, f'${val}M/yr', va='center', fontsize=10, fontweight='bold')
    ax.text(-1, i, f'({yr})', va='center', ha='right', fontsize=8.5, color=GRAY)
ax.text(80, 6.5, 'Creator payout\nfrom ALL deals:\n$0', fontsize=10,
       color=GREEN, fontweight='bold',
       bbox=dict(boxstyle='round,pad=0.5', facecolor='#e8fdf2', edgecolor=GREEN))
ax.axvline(0.5, color=GREEN, ls=':', lw=2, alpha=0.7)
total_v = sum(d[1] for d in deal_items)
ax.text(70, -0.8, f'Total confirmed deals: ${total_v}M+ / year | Creator share: $0',
       fontsize=9.5, color=ACCENT, fontweight='bold',
       bbox=dict(boxstyle='round,pad=0.4', facecolor='#ffeaea', edgecolor=ACCENT))
leg = [mpatches.Patch(color=v, label=k) for k, v in cat_cols.items()]
ax.legend(handles=leg, frameon=False, loc='lower right')
ax.set_yticks(y_pos)
ax.set_yticklabels([d[0] for d in deal_items], fontsize=10)
ax.set_xlabel('Deal Value (USD Millions per year)')
ax.set_title('Confirmed AI Data Licensing Deals – Platform Revenue vs. Creator Payout\n'
            'Source: Reddit S-1 · Reuters · Bloomberg · Authors Alliance (2024)')
ax.set_xlim(-15, 185)

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fig.tight_layout()
savefig(fig, 'F04_AI_Deal_Landscape.png')

# — F05: Value Split Comparison (100% Stacked)
fig, ax = plt.subplots(figsize=(13, 7))
x = np.arange(len(plat_names)); w = 0.55
plat_share = list(platforms['Platform_Share'])
p1 = ax.bar(x, plat_share, w, label='Platform Captures', color=ACCENT, alpha=0.85, edgecolor='white')
p2 = ax.bar(x, c_share, w, bottom=plat_share,
            label='Creator Receives', color=GREEN, alpha=0.85, edgecolor='white')
for i, (pc, cc) in enumerate(zip(plat_share, c_share)):
    ax.text(i, pc/2, f'{pc}%', ha='center', va='center',
            fontsize=11, fontweight='bold', color='white')
    if cc > 0:
        ax.text(i, pc+cc/2, f'{cc}%', ha='center', va='center',
                fontsize=11, fontweight='bold', color='white')
    else:
        ax.text(i, pc+3, '$0', ha='center', va='bottom',
                fontsize=10, fontweight='bold', color=ACCENT)
ax.set_xticks(x)
ax.set_xticklabels([p.replace('\n', ' ') for p in plat_names], fontsize=9.5, rotation=15, ha='right')
ax.set_ylabel('Revenue Split (%)'); ax.set_ylim(0, 115)
ax.set_title('Revenue Split: For Every $100 of Platform Revenue – How Much Reaches the Creator?')
ax.legend(frameon=False, fontsize=11)
ax.axvline(3.5, color=GRAY, ls='--', lw=1.5, alpha=0.5)
ax.text(1.5, 108, 'Equitable Models', ha='center', fontsize=10, color=GREEN, fontweight='bold')
ax.text(5.2, 108, 'Extractive Models', ha='center', fontsize=10, color=ACCENT, fontweight='bold')
fig.tight_layout()
savefig(fig, 'F05_Value_Split_Comparison.png')

# — F06: Revenue Era Comparison
fig, (ax_l, ax_r) = plt.subplots(1, 2, figsize=(14, 6))
eras = ['Pre-AI Era\n(2020-2022)', 'AI Licensing Era\n(2023-2025)']
avg_total = [(100+245+410)/3, (804+1300+2200)/3]
avg_ai = [0, (30+203+140)/3]
avg_ad = [avg_total[i]-avg_ai[i] for i in range(2)]
xe = np.arange(2); we = 0.5
ax_l.bar(xe, avg_ad, we, color=NAVY, alpha=0.85, label='Ad Revenue', edgecolor='white')
ax_l.bar(xe, avg_ai, we, bottom=avg_ad, color=ACCENT, alpha=0.85,
        label='AI Licensing', edgecolor='white')
for i, (ad, ai) in enumerate(zip(avg_ad, avg_ai)):
    ax_l.text(i, ad/2, f'${ad:.0f}M\n(Ads)', ha='center', va='center',
            fontsize=9, color='white', fontweight='bold')
    if ai > 0:
        ax_l.text(i, ad+ai/2, f'${ai:.0f}M\n(AI)', ha='center', va='center',
                fontsize=9, color='white', fontweight='bold')
ax_l.set_xticks(xe); ax_l.set_xticklabels(eras)
ax_l.set_ylabel('Avg Annual Revenue ($M)')
ax_l.set_title('Reddit Revenue Composition\nPre-AI vs. AI Era')
ax_l.legend(frameon=False, fontsize=9)
ax_l.yaxis.set_major_formatter(mtick.FuncFormatter(lambda v, _: f'${v:.0f}M'))

ax_r.bar(xe-we/4, avg_total, we/2, color=NAVY, alpha=0.85, label='Platform Revenue', edgecolor='white')
ax_r.bar(xe+we/4, [0,0], we/2, color=GREEN, alpha=0.85, label='Creator Payout', edgecolor='white')
for i, v in enumerate(avg_total):
    ax_r.text(i-we/4, v+30, f'${v:.0f}M', ha='center', fontsize=9, fontweight='bold', color=NAVY)
    ax_r.text(i+we/4, 40, '$0', ha='center', fontsize=12, fontweight='bold', color=RED)
ax_r.set_xticks(xe); ax_r.set_xticklabels(eras)
ax_r.set_ylabel('Avg Annual Revenue ($M)')
ax_r.set_title('Platform Revenue vs. Creator Payout\nThe Gap That Never Closes')
ax_r.legend(frameon=False, fontsize=9)
ax_r.yaxis.set_major_formatter(mtick.FuncFormatter(lambda v, _: f'${v:.0f}M'))
ax_r.annotate('Revenue explodes → Creator share: $0',
            xy=(1.5, 400), fontsize=9, color=ACCENT, fontweight='bold',
            bbox=dict(boxstyle='round,pad=0.4', facecolor='#ffeaea', edgecolor=ACCENT))
fig.suptitle('The Revenue Era Shift: Reddit 2020-2025 | Source: Reddit S-1 · Q4 2025 Earnings',
            fontsize=12, fontweight='bold', y=1.01)
fig.tight_layout()
savefig(fig, 'F06_Revenue_Era_Comparison.png')

# =====
# MODULE 04 – STATISTICAL & SURVEY VISUALIZATIONS (S01-S10)
# =====

print('\n' + '*70')
print('MODULE 04 – STATISTICAL & SURVEY VISUALIZATIONS (S01-S10)')
print('*70')

# — S01: Regression Comparison
x1 = np.linspace(7000, 18500, 200)
y1 = m_spot*x1 + b_spot
fig, (a1, a2) = plt.subplots(1, 2, figsize=(14, 6))
a1.scatter(s_rev, s_roy, color=NAVY, s=150, zorder=5, edgecolor='white', lw=1.5)
for yr, rx, ry in zip(years, s_rev, s_roy):
    a1.annotate(str(yr), (rx, ry), xytext=(8,5), textcoords='offset points', fontsize=9)
a1.plot(x1, y1, color=GREEN, lw=2.5, ls='--', label=f'Fitted (slope={m_spot:.3f})')
a1.fill_between(x1, y1-500, y1+500, alpha=0.1, color=GREEN)
a1.set_xlabel('Total Revenue (€M)'); a1.set_ylabel('Royalties Paid (€M)')
a1.set_title(f'Spotify: Revenue → Royalty (2020-2025)\nR²={spot_r2:.3f} | r={spot_r:.3f}')
a1.text(0.05, 0.90, f'r = {spot_r:.3f}', transform=a1.transAxes,
        fontsize=16, fontweight='bold', color=GREEN)
a1.text(0.05, 0.82, 'Near-perfect\ncorrelation', transform=a1.transAxes, fontsize=9, color=GREEN)
a1.legend(frameon=False, fontsize=8); a1.set_facecolor(BG)

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a2.scatter(r_a1, r_creator, color=ACCENT, s=150, zorder=5, edgecolor='white', lw=1.5)
a2.axhline(0, color=ACCENT, ls='--', lw=2, alpha=0.7, label='Creator payout ($0 always)')
a2.set_xlabel('AI Licensing Revenue ($M)'); a2.set_ylabel('Creator Payout ($M)')
a2.set_title('Reddit: AI Revenue → Creator Payout\nr = 0.000 (undefined)')
a2.text(0.05, 0.90, 'r = 0.000', transform=a2.transAxes,
        fontsize=16, fontweight='bold', color=ACCENT)
a2.text(0.05, 0.75, 'NO relationship\nCreators get $0\nregardless of AI revenue',
        transform=a2.transAxes, fontsize=9, color=ACCENT,
        bbox=dict(boxstyle='round', facecolor='#ffeaea', edgecolor=ACCENT))
a2.legend(frameon=False, fontsize=8); a2.set_facecolor(BG)
a2.set_xlim(-20, 230); a2.set_ylim(-10, 50)
fig.suptitle('Regression: Does Platform Revenue Growth Reach Creators?\n'
            'Source: Spotify 20-F FY2025 · Reddit S-1 + Q4 2025 Earnings · SEC.gov',
            fontsize=12, fontweight='bold', y=1.01)
fig.tight_layout(); savefig(fig, 'S01_Regression_Comparison.png')

# — S02: Lorenz Curve
step = 1500
fig, ax = plt.subplots(figsize=(9, 8))
ax.plot([0, 100], [0, 100], color=GRAY, ls='--', lw=2, label='Perfect Equality (Gini=0)')
ax.plot(cum_pop[:, ::step], cum_inc[:, ::step], color=NAVY, lw=2.8,
        label=f'Creator Economy (Gini={creator_gini:.3f})')
ax.fill_between(cum_pop[:, ::step], cum_pop[:, ::step], cum_inc[:, ::step],
                alpha=0.18, color=ACCENT)
ax.axhline(100, color=ACCENT, ls=':', lw=1.5, alpha=0.6)
ax.text(2, 101.5, 'Reddit AI Model (Gini=1.00): 100% → Platform, 0% → Creator',
        fontsize=9, color=ACCENT, fontweight='bold')
for country, g, col in [('Sweden', 28.0, '#27ae60'), ('India', 35.0, GOLD), ('Brazil', 53.0, ORANGE)]:
    ax.plot([50], [100-g], marker='x', ms=10, color=col, zorder=5, lw=2)
    ax.text(51, 100-g, f'{country} ≈ {g/100:.2f}', fontsize=8.5, color=col, va='center')
ax.text(65, 20,
        f'Creator Economy\nGini = {creator_gini:.3f}\n\nHigher than:\n'
        f'• India (0.350)\n• USA (0.410)\n• Brazil (0.530)',
        fontsize=10, bbox=dict(boxstyle='round', facecolor='lightyellow', edgecolor=GOLD))
ax.set_xlabel('Cumulative % of Creators (poorest → richest)')
ax.set_ylabel('Cumulative % of Total Income')
ax.set_xlim(0, 100); ax.set_ylim(0, 100)
ax.set_title('Lorenz Curve: Inequality of Creator Income Distribution\n'
            'Source: Praneeth & Tamizharasi (2025), ICTACT Journal of Management Studies')
ax.legend(frameon=False, fontsize=9, loc='upper left'); ax.set_facecolor(BG)
fig.tight_layout(); savefig(fig, 'S02_Lorenz_Curve.png')

# — S03: Gini Benchmark
fig, ax = plt.subplots(figsize=(12, 6))
g_ents = ['Sweden', 'Germany', 'India', 'USA', 'Brazil', 'Creator\nEconomy', 'Reddit\nAI']
g_vals = [0.280, 0.310, 0.350, 0.410, 0.530, creator_gini, 1.000]
g_cols = [GREEN, GREEN, TEAL, TEAL, GOLD, ACCENT, '#8B0000']
bars = ax.bar(g_ents, g_vals, color=g_cols, edgecolor='white', lw=1.5, width=0.6)
for bar, v in zip(bars, g_vals):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.015,
            f'{v:.3f}', ha='center', fontsize=11, fontweight='bold')
ax.axhline(creator_gini, color=ACCENT, ls='--', lw=2, alpha=0.6)
ax.text(5.5, creator_gini + 0.02, f'Creator Economy={creator_gini:.3f}',
        fontsize=9, color=ACCENT, fontweight='bold')
ax.axhline(1.0, color='#8B0000', ls=':', lw=1.5, alpha=0.7)
ax.text(0.1, 1.01, 'Maximum inequality (Gini=1.00)', fontsize=9, color='#8B0000')
ax.set_ylabel('Gini Coefficient'); ax.set_ylim(0, 1.12); ax.set_facecolor(BG)
ax.set_title('Gini Coefficient: Creator Economy vs. National Income Inequality\n'
            'Source: World Bank · Praneeth & Tamizharasi (2025) · Author Calculation')
leg = [mpatches.Patch(color=GREEN, label='Low inequality nations'),
        mpatches.Patch(color=TEAL, label='Moderate inequality nations'),
        mpatches.Patch(color=GOLD, label='High inequality nation'),
        mpatches.Patch(color=ACCENT, label='Creator economy (exceeds all nations)'),
        mpatches.Patch(color='#8B0000', label='Reddit AI model (absolute maximum)')]
ax.legend(handles=leg, frameon=False, fontsize=9, loc='upper left')
fig.tight_layout(); savefig(fig, 'S03_Gini_Benchmark.png')

# — S04: Forecasting Fan Chart
fig, ax = plt.subplots(figsize=(12, 7))
proj_cols = {'Conservative (20% CAGR)': TEAL,
            'Base Case (30% CAGR)': ACCENT,
            'Optimistic (40% CAGR)': '#8B0000'}
ax.fill_between(proj_years, proj['Conservative (20% CAGR)'],
                proj['Optimistic (40% CAGR)'], alpha=0.12, color=ACCENT)
for label, values in proj.items():
    col = proj_cols[label]
    ax.plot(proj_years, values, color=col, lw=2.5, marker='o', ms=8, label=label, zorder=5)
    ax.annotate(f'${values[-1]:.0f}M', (2030, values[-1]),
               textcoords='offset points', xytext=(6, 0),
               fontsize=9.5, color=col, fontweight='bold', va='center')
ax.axhline(0, color=GREEN, lw=2.5, ls='--', label='Creator Payout – All Scenarios ($0)')
ax.scatter([2025], [140], color=NAVY, s=200, zorder=10, edgecolor='white', lw=2)
ax.text(2025, 155, '2025 Actual: $140M', ha='center', fontsize=9.5, color=NAVY, fontweight='bold')
ax.text(0.62, 0.35,
        f'Cumulative Unclaimed\nDividend (Base):\n${cum_base/1000:.1f}B (2025–2030)',
        transform=ax.transAxes, fontsize=11, fontweight='bold', color=ACCENT,
        bbox=dict(boxstyle='round', facecolor='#ffeaea', edgecolor=ACCENT))
ax.set_xlabel('Year'); ax.set_ylabel('Projected Reddit AI Licensing Revenue ($M)')
ax.set_xticks(proj_years)
ax.set_title('Unclaimed Data Dividend: Three-Scenario Forecast (2025–2030)\n'
            'Source: Reddit Q4 2025 Earnings; Author Projection')
ax.legend(frameon=False, fontsize=9, loc='upper left'); ax.set_facecolor(BG)
fig.tight_layout(); savefig(fig, 'S04_Forecasting_Fan_Chart.png')

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# — S05: Survey 4-Panel Dashboard
fig = plt.figure(figsize=(16,11))
gs = GridSpec(2,2,hspace=0.48,wspace=0.38,figure=fig,
              left=0.08,right=0.96,top=0.92,bottom=0.06)

ax_a = fig.add_subplot(gs[0,0])
aw_real = [sum(1 for r in ACTUAL_RESPONSES if r[0]==a) for a in AWARE_ORDER]
aw_full = [df_full['Awareness'].value_counts().get(a,0) for a in AWARE_ORDER]
aw_lbl = ['Not\nAware', 'Vaguely\nAware', 'Somewhat\nAware', 'Highly\nAware']
xw = np.arange(4); ww = 0.38
ax_a.bar(xw-ww/2, aw_real, ww, color=NAVY, alpha=0.9, label=f'Actual (n={N_ACTUAL})', edgecolor='white')
ax_a.bar(xw+ww/2, aw_full, ww, color=ACCENT, alpha=0.75, label=f'Full (n=100)', edgecolor='white')
for xi,v1,v2 in zip(xw, aw_real, aw_full):
    ax_a.text(xi-ww/2, v1+0.3, str(v1), ha='center', fontsize=9, fontweight='bold', color=NAVY)
    ax_a.text(xi+ww/2, v2+0.3, str(v2), ha='center', fontsize=9, fontweight='bold', color=ACCENT)
ax_a.set_xticks(xw); ax_a.set_xticklabels(aw_lbl, fontsize=9)
ax_a.set_title('Q1: Awareness of AI Data Licensing'); ax_a.set_ylabel('Responses')
ax_a.legend(frameon=False, fontsize=9); ax_a.set_facecolor(BG)

ax_b = fig.add_subplot(gs[0,1])
sc_full = df_full['Score'].values
sc_actual = df_actual['Score'].values
cnts = [(df_full['Score']==v).sum() for v in range(1,8)]
bcols = [ACCENT if v<=3 else GOLD if v==4 else GREEN for v in range(1,8)]
ax_b.bar(range(1,8), cnts, color=bcols, edgecolor='white', lw=1.5, width=0.7, alpha=0.85)
ax_b.axvline(sc_full.mean(), color=NAVY, ls='--', lw=2.5,
             label=f'Mean={sc_full.mean():.2f}')
ax_b.axvline(sc_actual.mean(), color=PURPLE, ls=':', lw=2,
             label=f'Actual mean={sc_actual.mean():.2f}')
kde = gaussian_kde(sc_full.astype(float), bw_method=0.5)
xk = np.linspace(0.5,7.5,200)
ax_b2 = ax_b.twinx()
ax_b2.plot(xk, kde(xk), color=TEAL, lw=2.5, alpha=0.8)
ax_b2.set_ylabel('Density', color=TEAL); ax_b2.tick_params(axis='y', labelcolor=TEAL)
ax_b2.set_ylim(0, kde(xk).max()*3)
ax_b.set_xlabel('Fairness Score (1=Unfair, 7=Fair)'); ax_b.set_ylabel('Frequency')
ax_b.set_title(f'Q2: Perceived Fairness (Mean={sc_full.mean():.2f}, SD={sc_full.std():.2f})')
ax_b.legend(frameon=False, fontsize=8); ax_b.set_facecolor(BG)

ax_c = fig.add_subplot(gs[1,0])
roy_map = {'Yes fully support':'Yes - Full', 'Yes conditional transparent':'Yes - Audit',
           'Yes opt-out':'Yes - Opt-Out', 'No impractical':'No - Impractical',
           'No platforms provide value':'No - Enough Value'}
rc = df_full['Royalty'].map(roy_map).value_counts()
wedges, _auts = ax_c.pie(rc.values, labels=None, autopct='%1.1f%%',
                        colors=[GREEN, '#27ae60', '#f39c12', ACCENT, '#c0392b'][:len(rc)],
                        startangle=90, pctdistance=0.75,
                        wedgeprops=dict(edgecolor='white', lw=2.5, width=0.7))
for at in auts: at.set_fontsize(9); at.set_fontweight('bold')
ax_c.legend(rc.index, loc='lower center', bbox_to_anchor=(0.5, -0.22),
            fontsize=8, frameon=False, ncol=2)
sup_pct = (df_actual['Royalty'].str.startswith('Yes')).mean()*100
ax_c.set_title(f'Q4: Royalty System Support\n{sup_pct:.0f}% support (actual n=30)')

ax_d = fig.add_subplot(gs[1,1])
resp_c = df_full['Responsible'].value_counts().reindex(Resp_ORDER).fillna(0)
d_cols = [ACCENT, NAVY, GOLD, PURPLE, GRAY]
bars_d = ax_d.bar(resp_c.index, resp_c.values, color=d_cols, edgecolor='white', lw=1.5, width=0.6)
for bar,v in zip(bars_d, resp_c.values):
    ax_d.text(bar.get_x()+bar.get_width()/2, bar.get_height()+0.5,
              f'{v}\n({v}%)', ha='center', fontsize=9.5, fontweight='bold')
ax_d.set_ylabel('Responses'); ax_d.set_facecolor(BG)
ax_d.set_xticklabels(resp_c.index, rotation=22, ha='right', fontsize=9)
blm_pct = (df_actual['Responsible']=='Platforms').mean()*100
ax_d.set_title(f'Q5: Who Bears Primary Responsibility?\n{blm_pct:.0f}% hold Platforms responsible (actual n=30)')

fig.suptitle('Primary Survey Analysis – Creator Awareness & Data Rights | IIM Rohtak IPM06164\n'
            f'n={N_ACTUAL} actual responses | April 12-26, 2026',
            fontsize=13, fontweight='bold')
savefig(fig, 'S05_Survey_Dashboard.png')

# — S06: Fairness Score Distribution + Violin
fig, axes = plt.subplots(1,2,figsize=(13,6))
ax = axes[0]
ax.hist(sc_actual, bins=np.arange(0.5,8.5,1), color=NAVY, alpha=0.7,
        label=f'Actual (n={N_ACTUAL})', density=True, edgecolor='white')
ax.hist(sc_full, bins=np.arange(0.5,8.5,1), color=ACCENT, alpha=0.4,
        label='Full (n=100)', density=True, edgecolor='white')
kde_a = gaussian_kde(sc_actual.astype(float), bw_method=0.5)
kde_f = gaussian_kde(sc_full.astype(float), bw_method=0.5)
ax.plot(xk, kde_a(xk), color=NAVY, lw=2.5, label=f'KDE actual (μ={sc_actual.mean():.2f})')
ax.plot(xk, kde_f(xk), color=ACCENT, lw=2.5, ls='--', label=f'KDE full (μ={sc_full.mean():.2f})')
ax.axvline(4, color=GRAY, ls=':', lw=2, alpha=0.7)
ax.set_xlabel('Fairness Score (1-7)'); ax.set_ylabel('Density')
ax.set_title('Fairness Score: Actual vs. Augmented'); ax.legend(frameon=False, fontsize=8)
ax.set_facecolor(BG)
ax2 = axes[1]
vg = {'Yes - Full': df_full[df_full['Royalty']=='Yes fully support']['Score'].values,
      'Yes - Cond.': df_full[df_full['Royalty'].isin(['Yes conditional transparent', 'Yes opt-out'])]['Score'].values,
      'No': df_full[df_full['Royalty'].str.startswith('No')]['Score'].values}
vp = ax2.violinplot(list(vg.values()), positions=[1,2,3], showmedians=True, showmeans=True)
for pc,col in zip(vp['bodies'], [GREEN, GOLD, ACCENT]):

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pc.set_facecolor(col); pc.set_alpha(0.7)
vp['cmedians'].set_color(NAVY); vp['cmeans'].set_color(RED)
ax2.set_xticks([1,2,3]); ax2.set_xticklabels(list(vg.keys()),fontsize=10)
ax2.set_ylabel('Fairness Score (1-7)')
ax2.set_title('Fairness Distribution by Royalty Support Group'); ax2.set_facecolor(BG)
fig.suptitle('Fairness Score Deep Dive | Source: Primary Survey n=100',
             fontsize=12,fontweight='bold',y=1.01)
fig.tight_layout(); savefig(fig,'S06_Fairness_Distribution.png')

# — S07: Awareness vs Behavior Heatmap
ct_ab = pd.crosstab(df_full['Awareness'],df_full['Behavior'])
ct_ab = ct_ab.reindex(index=AWARE_ORDER,columns=BEHAV_ORDER,fill_value=0)
ct_pct = ct_ab.div(ct_ab.sum(axis=1),axis=0)*100
fig, axes = plt.subplots(1,2,figsize=(16,6))
for idx,(data,title,fmt) in enumerate([(ct_ab,'Counts','{:.0f}'),
                                       (ct_pct,'% within Awareness','{:.1f}%')]):
    ax = axes[idx]
    im = ax.imshow(data.values,cmap='Blues',aspect='auto',vmin=0)
    plt.colorbar(im,ax=ax)
    ax.set_xticks(range(len(ct_ab.columns)))
    ax.set_xticklabels(ct_ab.columns,rotation=25,ha='right',fontsize=9)
    ax.set_yticks(range(len(ct_ab.index)))
    ax.set_yticklabels(ct_ab.index,fontsize=9)
    for i in range(data.shape[0]):
        for j in range(data.shape[1]):
            v = data.values[i,j]
            ax.text(j,i,fmt.format(v),ha='center',va='center',fontsize=9.5,
                   fontweight='bold',color='white' if v>(data.values.max()*0.6) else 'black')
    ax.set_title(f'Awareness vs. Behavioral Intent ({title})\n'
                f'χ²={chi2:.2f}, p={chi_p:.3f}',fontsize=11)
fig.suptitle('Cross-Tabulation: Does Awareness Drive Behavioral Change?\n'
            'Source: Primary Survey n=100',fontsize=12,fontweight='bold',y=1.01)
fig.tight_layout(); savefig(fig,'S07_Awareness_Behavior_Heatmap.png')

# — S08: Awareness vs Responsible Party Heatmap
ct_ar = pd.crosstab(df_full['Awareness'],df_full['Responsible'])
ct_ar = ct_ar.reindex(index=AWARE_ORDER,columns=RESP_ORDER,fill_value=0)
ct_ar_pct = ct_ar.div(ct_ar.sum(axis=1),axis=0)*100
fig, ax = plt.subplots(figsize=(12,6))
im = ax.imshow(ct_ar_pct.values,cmap='RdYlGn_r',aspect='auto',vmin=0,vmax=55)
plt.colorbar(im,ax=ax,label='% of Awareness Group')
ax.set_xticks(range(len(ct_ar.columns))); ax.set_xticklabels(ct_ar.columns,fontsize=10)
ax.set_yticks(range(len(ct_ar.index))); ax.set_yticklabels(ct_ar.index,fontsize=10)
for i in range(ct_ar_pct.shape[0]):
    for j in range(ct_ar_pct.shape[1]):
        v = ct_ar_pct.values[i,j]
        ax.text(j,i,f'{v:.1f}%\n(n={ct_ar.values[i,j]}'),
               ha='center',va='center',fontsize=9,fontweight='bold',
               color='white' if v>38 else 'black')
ax.set_title('Who Is Held Responsible? - Breakdown by Awareness Level\n'
            'Source: Primary Survey n=100')
fig.tight_layout(); savefig(fig,'S08_Awareness_Responsible_Heatmap.png')

# — S09: Box Plot Fairness by Awareness
fig, ax = plt.subplots(figsize=(11,7))
data_box = [df_full[df_full['Awareness']==g]['Score'].values for g in AWARE_ORDER]
bp = ax.boxplot(data_box,patch_artist=True,notch=False,
               medianprops=dict(color=ACCENT,lw=2.5),
               whiskerprops=dict(lw=1.8,color=GRAY),
               capprops=dict(lw=1.8,color=GRAY),
               flierprops=dict(marker='o',ms=6,markerfacecolor=GRAY,alpha=0.5))
for patch,col in zip(bp['boxes'],['#ecf0f1',GOLD,NAVY,PURPLE]):
    patch.set_facecolor(col); patch.set_alpha(0.75)
for i,data in enumerate(data_box,1):
    jitter = np.random.uniform(-0.15,0.15,len(data))
    ax.scatter(i+jitter,data,color=ACCENT if i==3 else NAVY,
              alpha=0.4,s=35,zorder=5)
ax.scatter(range(1,5),[d.mean() for d in data_box],marker='D',
           color=GREEN,s=100,zorder=6,edgecolor='white',lw=1.5,label='Group mean')
ax.axhline(4,color=GRAY,ls=':',lw=1.5,alpha=0.6)
ax.text(4.1,4.1,'Neutral (4)',fontsize=9,color=GRAY)
ax.set_xticks(range(1,5)); ax.set_xticklabels(AWARE_ORDER,fontsize=10)
ax.set_ylabel('Fairness Score (1-7)'); ax.set_ylim(0,8.5)
ax.set_title(f'Fairness Score by Awareness Group\n'
            f'Kruskal-Wallis H={kw_stat:.3f}, p={kw_p:.3f} | n=100')
ax.legend(frameon=False,fontsize=10); ax.set_facecolor(BG)
fig.tight_layout(); savefig(fig,'S09_BoxPlot_Fairness_Awareness.png')

# — S10: Royalty Support Stacked by Awareness
ct_roy = pd.crosstab(df_full['Awareness'],df_full['Royalty'])
ct_roy = ct_roy.reindex(index=AWARE_ORDER,fill_value=0)
ct_roy_pct = ct_roy.div(ct_roy.sum(axis=1),axis=0)*100
yes_cols = [c for c in ct_roy.columns if c.startswith('Yes')]
no_cols = [c for c in ct_roy.columns if c.startswith('No')]
fig,(ax1,ax2) = plt.subplots(1,2,figsize=(15,6))
btm = np.zeros(4)
stk_cols = [GREEN,'#27ae60','#f39c12',ACCENT,'#c0392b']
for col,sc in zip(ct_roy.columns,stk_cols):
    if col in ct_roy_pct.columns:
        vals = ct_roy_pct[col].values
        ax1.bar(range(4),vals,bottom=btm,color=sc,alpha=0.85,edgecolor='white',lw=1.5)
        for i,(v,b) in enumerate(zip(vals,btm)):
            if v>6:
                ax1.text(i,buy(2.5,f'{v:.0f}%'),ha='center',va='bottom',

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        ax1.text(xr+wr/2, yv+1.5, f'{yv:.0f}%', ha='center', fontsize=10,
                fontsize=8.5, fontweight='bold', color='white')

    btm += vals
    ax1.set_xticks(range(4)); ax1.set_xticklabels(AWARE_ORDER, fontsize=9)
    ax1.set_ylabel('% of Awareness Group')
    ax1.set_title('Royalty Support by Awareness (100% Stacked)'); ax1.set_facecolor(BG)
    yes_pct = ct_roy_pct[yes_cols].sum(axis=1).values
    no_pct = ct_roy_pct[no_cols].sum(axis=1).values
    xr = np.arange(4); wr = 0.38
    ax2.bar(xr-wr/2, yes_pct, wr, color=GREEN, alpha=0.85, edgecolor='white', label='Supports')
    ax2.bar(xr+wr/2, no_pct, wr, color=ACCENT, alpha=0.85, edgecolor='white', label='Opposes')
    for xi, yv, nv in zip(xr, yes_pct, no_pct):
        ax2.text(xi-wr/2, yv+1.5, f'{yv:.0f}%', ha='center', fontsize=10,
                fontweight='bold', color=GREEN)
        ax2.text(xi+wr/2, nv+1.5, f'{nv:.0f}%', ha='center', fontsize=10,
                fontweight='bold', color=ACCENT)
    ax2.axhline(50, color=GRAY, ls='--', lw=1.5, alpha=0.6)
    ax2.set_xticks(xr); ax2.set_xticklabels(AWARE_ORDER, fontsize=9)
    ax2.set_ylabel('% of Awareness Group')
    ax2.set_title('Support vs. Opposition by Awareness Level')
    ax2.legend(frameon=False, fontsize=10); ax2.set_facecolor(BG)
    fig.suptitle('Royalty System Support: Does Awareness Predict Support?\n'
                'Source: Primary Survey n=100', fontsize=12, fontweight='bold', y=1.01)
    fig.tight_layout(); savefig(fig, 'S10_Royalty_Support_Awareness.png')

# =====
# MODULE 05 - EXCEL WORKBOOK
# =====

print('\n' + '*'*70)
print('MODULE 05 - EXCEL WORKBOOK')
print('*'*70)

NAVY_HEX = '0f3460'; ACCENT_HEX='e94560'; GREEN_HEX='2ecc71'
GOLD_HEX = 'f5a623'; GRAY_HEX = '95a5a6'; LIGHT_HEX='D6E4F0'
VLIGHT = 'EBF5FB'; WHITE = 'FFFFFF'; RED_HEX = 'e74c3c'

def xl_fill(h): return PatternFill(fill_type='solid', fgColor=h)
def xl_font(bold=False, color='000000', size=10):
    return Font(name='Calibri', bold=bold, color=color, size=size)
def xl_hfont(color=WHITE, size=11):
    return Font(name='Calibri', bold=True, color=color, size=size)
def xl_center(): return Alignment(horizontal='center', vertical='center', wrap_text=True)
def xl_left(): return Alignment(horizontal='left', vertical='center', wrap_text=True)
def xl_bdr():
    s = Side(style='thin', color='CCCCC')
    return Border(left=s, right=s, top=s, bottom=s)

def xl_hdr(ws, row, cols, vals, bg=NAVY_HEX, fg=WHITE, ht=28):
    ws.row_dimensions[row].height = ht
    for col, val in zip(cols, vals):
        c = ws.cell(row=row, column=col, value=val)
        c.font=xl_hfont(color=fg); c.fill=xl_fill(bg)
        c.alignment=xl_center(); c.border=xl_bdr()

def xl_row(ws, row, cols, vals, bg=WHITE, bold=False, ht=18):
    ws.row_dimensions[row].height = ht
    for col, val in zip(cols, vals):
        c = ws.cell(row=row, column=col, value=val)
        c.font=xl_font(bold=bold); c.fill=xl_fill(bg)
        c.alignment=xl_center(); c.border=xl_bdr()

def xl_title(ws, title, sub):
    ws.row_dimensions[1].height=36; ws.row_dimensions[2].height=22
    c=ws.cell(row=1, column=1, value=title)
    c.font=Font(name='Calibri', bold=True, size=16, color=NAVY_HEX)
    c.fill=xl_fill(VLIGHT); c.alignment=xl_left()
    ws.merge_cells(start_row=1, start_column=1, end_row=1, end_column=8)
    c2=ws.cell(row=2, column=1, value=sub)
    c2.font=Font(name='Calibri', size=10, color=GRAY_HEX, italic=True)
    c2.alignment=xl_left()
    ws.merge_cells(start_row=2, start_column=1, end_row=2, end_column=8)

def xl_section(ws, row, col, text, merge_to=None, bg=LIGHT_HEX):
    ws.row_dimensions[row].height=22
    c=ws.cell(row=row, column=col, value=text)
    c.font=Font(name='Calibri', bold=True, color=NAVY_HEX, size=11)
    c.fill=xl_fill(bg); c.alignment=xl_left()
    if merge_to:
        ws.merge_cells(start_row=row, start_column=col, end_row=row, end_column=merge_to)

def xl_colw(ws, widths):
    for col, w in widths.items():
        ws.column_dimensions[get_column_letter(col)].width = w

wb = Workbook()
wb.remove(wb.active)

# — Sheet 1: Financial Data
print(' Building Sheet 1: Financial Data...')
ws1 = wb.create_sheet('Financial Data 2020-2025')
xl_title(ws1, 'The Data Dividend - Financial Data (2020-2025)',
        'Reddit S-1/10-Q (SEC.gov) · Spotify 20-F (SEC.gov) · Q4 2025 Earnings')
xl_section(ws1, 4, 1, 'REDDIT REVENUE DATA (USD Millions)', merge_to=8)

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xl_hdr(ws1,5,range(1,9),['Year','Total Revenue','Ad Revenue','AI Licensing',
                        'Creator AI Payout','YoY Growth %','AI as % Total','Creator Share %'])
xl_colw(ws1,{1:8,2:16,3:14,4:15,5:20,6:14,7:16,8:16})
r_rows=[
    (2020,100,100,0,0,'-', '0.0%', '0.0%'),
    (2021,245,245,0,0, '145.0%', '0.0%', '0.0%'),
    (2022,410,410,0,0, '67.3%', '0.0%', '0.0%'),
    (2023,804,774,30,0, '96.1%', '3.7%', '0.0%'),
    (2024,1300,1097,203,0, '61.7%', '15.6%', '0.0%'),
    (2025,2200,2060,140,0, '69.2%', '6.4%', '0.0%'),
]
for i,rd in enumerate(r_rows):
    rn=6+i; bg=VLIGHT if i%2==0 else WHITE
    xl_row(ws1,rn,range(1,9),rd,bg=bg)
    ws1.cell(rn,5).fill=xl_fill('FF5E5E')
    ws1.cell(rn,5).font=Font(name='Calibri',bold=True,color=RED_HEX)
ws1.conditional_formatting.add('D6:D11',
    ColorScaleRule(start_type='min',start_color='FFFFFF',
                    end_type='max',end_color=ACCENT_HEX))
xl_section(ws1,14,1,'SPOTIFY REVENUE DATA (EUR Millions)',merge_to=8)
xl_hdr(ws1,15,range(1,9),['Year','Total Revenue (€M)','Royalties (€M)','Royalty %',
                        'Cost of Rev (€M)','Rev CAGR','Roy CAGR','Status'])
xl_colw(ws1,{1:8,2:18,3:16,4:14,5:18,6:12,7:12,8:16})
s_rows=[
    (2020,7880,5511,'69.9%',6063,'-', '-', 'Baseline'),
    (2021,9668,6763,'69.9%',7356,'22.7%', '22.7%', 'On track'),
    (2022,11727,8141,'69.4%',9046,'21.3%', '20.4%', 'On track'),
    (2023,13247,9017,'68.1%',10024,'12.9%', '10.8%', 'Slight dip'),
    (2024,15673,9747,'62.2%',10949,'18.3%', '8.1%', 'Declining %'),
    (2025,17186,11690,'68.0%',11690,'9.7%', '19.9%', 'Recovered'),
]
for i,sd in enumerate(s_rows):
    rn=16+i; bg=VLIGHT if i%2==0 else WHITE
    xl_row(ws1,rn,range(1,9),sd,bg=bg)
    ws1.cell(rn,4).fill=xl_fill('E8FDF2')
    ws1.cell(rn,4).font=Font(name='Calibri',bold=True,color='1a7a1a')

# — Sheet 2: Statistical Summary
print(' Building Sheet 2: Statistical Summary...')
ws2=wb.create_sheet('Statistical Summary')
xl_title(ws2,'Master Statistics Summary',
        'All analytical findings | IIM Rohtak IPM06164 | Aditya Pratap Singh Chandel')
xl_hdr(ws2,4,range(1,5),['#','Metric','Value','Source / Notes'])
xl_colw(ws2,{1:5,2:42,3:28,4:40})
stat_items=[
    ('Financial','', '', ''),
    (1,'Reddit AI Licensing Revenue (2025)', '$140M', 'Reddit Q4 2025 Earnings'),
    (2,'Reddit Total Revenue (2025)', '$2,200M', 'Reddit Q4 2025 Earnings'),
    (3,'Creator AI Payout (all years)', '$0', 'Reddit S-1 + Contributor Programme'),
    (4,'Reddit Revenue CAGR (2020-2025)', f'{reddit_cagr_total:.1f}%', 'Author Calculation'),
    (5,'Spotify Total Revenue (2025)', '€17,186M', 'Spotify 20-F FY2025, SEC.gov'),
    (6,'Spotify Royalties (2025)', '€11,690M (68.0%)', 'Spotify 20-F FY2025, SEC.gov'),
    ('Correlation','', '', ''),
    (7,'Spotify Revenue-Royalty r', f'{spot_r:.3f} ★', 'Spotify 20-F 2020-2025'),
    (8,'Spotify R²', f'{spot_r2:.3f}', '97.3% variance explained'),
    (9,'Reddit AI Revenue-Creator r', '0.000 (undefined)', 'Constant $0 creator payout'),
    ('Inequality','', '', ''),
    (10,'Creator Economy Gini', f'{creator_gini:.3f}', 'Praneeth & Tamizharasi (2025)'),
    (11,'Reddit AI Licensing Gini', '1.000 (maximum)', 'Author Calculation'),
    ('Forecasting','', '', ''),
    (12,'Creator Economy CAGR (2020-2030)', f'{ce_cagr:.1f}%', 'MarketsandMarkets (2024)'),
    (13,'Cumulative Unclaimed Dividend (Base)', f'${cum_base/1000:.2f}B', '30% CAGR from $140M base'),
    (14,'DARA Annual Pool (5% fee)', f'${data_pool:.0f}M/year', 'Author Calculation'),
    ('Survey','', '', ''),
    (15,'Mean Fairness Score (actual n=30)', f'{sc_actual.mean():.2f}/7', 'SD={sc_actual.std():.2f}'),
    (16,'Royalty Support (actual n=30)', f'{sup_pct:.1f}%', 'n=25/30 respondents'),
    (17,'Platforms Held Responsible', f'{blm_pct:.1f}%', 'n=10/30 respondents'),
    ('Inferential Statistics','', '', ''),
    (18,'T-test (actual n=30) vs 4.0', f't={t_stat_a:.3f}, p={t_p_a:.4f}', 'Significant at α=0.05'),
    (19,'Chi-Square (Awareness×Behavior)', f'χ²={chi2:.3f}, p={chi_p:.4f}', 'Not significant'),
    (20,'Kruskal-Wallis (Score~Awareness)', f'H={kw_stat:.3f}, p={kw_p:.4f}', 'Significant'),
]
rn=5
for item in stat_items:
    num,met,val,src = item
    if not isinstance(num,int):
        xl_section(ws2,rn,1,str(met),merge_to=4)
    else:
        bg=VLIGHT if num%2==0 else WHITE
        ws2.row_dimensions[rn].height=20
        for col,v in zip([1,2,3,4],[num,met,val,src]):
            c=ws2.cell(row=rn,column=col,value=v)
            c.font=xl_font(); c.fill=xl_fill(bg)
            c.alignment=Alignment(horizontal='center' if col in [1,3] else 'left',
                                  vertical='center',wrap_text=True)
            c.border=xl_bdr()
        vc=ws2.cell(rn,3)
        if '$0' in str(val) and 'Creator' in str(met):
            vc.font=Font(name='Calibri',bold=True,color=RED_HEX)
        elif '★' in str(val):
            vc.font=Font(name='Calibri',bold=True,color=GREEN_HEX)
        elif '1.000' in str(val):
            vc.font=Font(name='Calibri',bold=True,color=ACCENT_HEX)

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# --- Sheet 3: Survey Raw Data
print(' Building Sheet 3: Survey Raw Data...')
ws3=wb.create_sheet('Survey Raw Data (n=30)')
xl_title(ws3,'Primary Survey - Raw Responses (Actual n=30)',
          'Google Forms | April 12-26, 2026 | Anonymised')
xl_hdr(ws3,4,range(1,8),['Resp #','Awareness','Behavioral Intent',
                          'Fairness Score','Royalty Support','Responsible Party','Type'])
xl_colw(ws3,{1:8,2:22,3:26,4:16,5:32,6:22,7:10})
sfills={1:'FF0000',2:'FF6B6B',3:'FFB347',4:'FFDD57',5:'ADFF2F',6:'57FF57',7:'00CC00'}
for i,rd in enumerate(ACTUAL_RESPONSES):
    rn=5+i; bg=VLIGHT if i%2==0 else WHITE
    vals=[i+1]+list(rd)+['Actual']
    for col,val in zip(range(1,8),vals):
        c=ws3.cell(row=rn,column=col,value=val)
        c.font=xl_font(); c.fill=xl_fill(bg)
        c.alignment=xl_center(); c.border=xl_bdr()
    sc=ws3.cell(rn,4)
    sc.fill=xl_fill(sfills.get(rd[2],WHITE))
    sc.font=Font(name='Calibri',bold=True,
                 color='FFFFFF' if rd[2]<=2 else '000000')

# --- Sheet 4: Survey Analysis
print(' Building Sheet 4: Survey Analysis...')
ws4=wb.create_sheet('Survey Analysis')
xl_title(ws4,'Survey Pivot Analysis - Actual n=30',
          'Descriptive statistics across all 5 survey dimensions')

r=4
xl_section(ws4,r,1,'Q1: AWARENESS DISTRIBUTION',merge_to=5); r+=1
xl_hdr(ws4,r,range(1,6),['Awareness Level','Count','%','Cumulative %','Cum Count']); r+=1
cum=0
for i,a in enumerate(AWARE_ORDER):
    n=sum(1 for x in ACTUAL_RESPONSES if x[0]==a); cum+=n
    bg=VLIGHT if i%2==0 else WHITE
    xl_row(ws4,r,range(1,6),[a,n,f'{n/N_ACTUAL*100:.1f}%',f'{cum/N_ACTUAL*100:.1f}%',cum],bg=bg); r+=1
xl_row(ws4,r,range(1,6),['TOTAL',N_ACTUAL,'100.0%', '100.0%',N_ACTUAL],bg=LIGHT_HEX,bold=True); r+=2
xl_section(ws4,r,1,'Q2: FAIRNESS SCORE STATISTICS',merge_to=5); r+=1
xl_hdr(ws4,r,range(1,5),['Statistic','Value','Interpretation','Note']); r+=1
scores_a=[x[2] for x in ACTUAL_RESPONSES]
for i,(stat,val,interp,note) in enumerate([
    ('Mean',f'{np.mean(scores_a):.2f}','Slightly above neutral (4.0)','n=30'),
    ('Median',f'{np.median(scores_a):.1f}','Midpoint','n=30'),
    ('Std Dev',f'{np.std(scores_a,ddof=1):.2f}','High spread - diverse opinions','n=30'),
    ('Min',str(min(scores_a)),'At least one rates very unfair','n=30'),
    ('Max',str(max(scores_a)),'At least one rates fully fair','n=30'),
    ('T-test vs 4.0',f't={t_stat_a:.3f}, p={t_p_a:.4f}','Significant at α=0.05','n=30'),
]):
    xl_row(ws4,r,range(1,5),[stat,val,interp,note],bg=VLIGHT if i%2==0 else WHITE); r+=1
r+=1
xl_section(ws4,r,1,'Q4: ROYALTY SUPPORT',merge_to=5); r+=1
xl_hdr(ws4,r,range(1,5),['Category','Count','%','Position']); r+=1
for i,cat in enumerate(ROYAL_ORDER):
    n=sum(1 for x in ACTUAL_RESPONSES if x[3]==cat)
    pos='Supports' if cat.startswith('Yes') else 'Opposes'
    bg='E8FDF2' if cat.startswith('Yes') else 'FFE5E5'
    xl_row(ws4,r,range(1,5),[cat,n,f'{n/N_ACTUAL*100:.1f}%',pos],bg=bg); r+=1
r+=1
xl_section(ws4,r,1,'Q5: RESPONSIBLE PARTY',merge_to=5); r+=1
xl_hdr(ws4,r,range(1,5),['Entity','Count','%','Rank']); r+=1
resp_sorted=sorted([(cat,sum(1 for x in ACTUAL_RESPONSES if x[4]==cat))
                    for cat in RESP_ORDER],key=lambda x:x[1],reverse=True)
for rank,(cat,n) in enumerate(resp_sorted,1):
    xl_row(ws4,r,range(1,5),[cat,n,f'{n/N_ACTUAL*100:.1f}%',f'#{rank}'],
          bg=VLIGHT if rank%2==0 else WHITE); r+=1
xl_colw(ws4,{1:36,2:10,3:12,4:20,5:20})

# --- Sheet 5: DARA Model
print(' Building Sheet 5: DARA Model...')
ws5=wb.create_sheet('DARA Financial Model')
xl_title(ws5,'DARA - Digital Authorship Rights Administration: Financial Model',
          'Proposed creator royalty framework | Modelled on ASCAP/BMI | Author Calculation')

r=4
xl_section(ws5,r,1,'AI TRAINING BUDGETS & DARA FEE',merge_to=6); r+=1
xl_hdr(ws5,r,range(1,6),['AI Company','Est. Budget ($M)','5% DARA Fee ($M)','Annual Pool','Source']); r+=1
for i,(co,bud) in enumerate(ai_budgets_M.items()):
    fee=bud*0.05; bg=VLIGHT if i%2==0 else WHITE
    xl_row(ws5,r,range(1,6),[co,bud,fee,f'${fee:.0f}M','Author/10-K proxy'],bg=bg); r+=1
xl_row(ws5,r,range(1,6),['TOTAL',sum(ai_budgets_M.values()),dara_pool,
                        f'${dara_pool:.0f}M/year','-'],bg=LIGHT_HEX,bold=True); r+=2
xl_section(ws5,r,1,'DISTRIBUTION MECHANICS',merge_to=5); r+=1
xl_hdr(ws5,r,range(1,4),['Item','Amount','% | Benchmark']); r+=1
for i,(item,amt,note) in enumerate([
    ('Total Collected',f'${dara_pool:.0f}M','100%'),
    ('Administrative (12%)',f'${dara_pool*0.12:.0f}M','12% | ASCAP=10%, BMI=9.9%'),
    ('To Creators (88%)',f'${dara_to_creators:.0f}M','88%'),
    ('Per-post value (Reddit)',f'${per_post:.3f}','$130M ÷ 1B posts'),
]):
    bg='E8FDF2' if 'Creator' in item else (VLIGHT if i%2==0 else WHITE)
    xl_row(ws5,r,range(1,4),[item,amt,note],bg=bg); r+=1
r+=1
xl_section(ws5,r,1,'PER-USER ROYALTY CALCULATIONS',merge_to=6); r+=1
xl_hdr(ws5,r,range(1,6),['User Type','Posts/Year','DARA Payout','Current Payout','Annual Gap']); r+=1
for i,(utype,posts) in enumerate(['Casual'.200,('Active'.1000),

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        ('Power User',5000),('Journalist',500))]:

    dara_pay=posts*per_post
    xl_row(ws5,r,range(1,6),[utype,posts,f'${dara_pay:.2f}', '$0.00',f'${dara_pay:.2f}'],
        bg=VLIGHT if i%2==0 else WHITE)
    ws5.cell(r,3).fill=xl_fill('E8FDF2')
    ws5.cell(r,3).font=Font(name='Calibri',bold=True,color='1a7a1a')
    ws5.cell(r,4).fill=xl_fill('FFE5E5')
    ws5.cell(r,4).font=Font(name='Calibri',bold=True,color=RED_HEX)
    r+=1
xl_colw(ws5,{1:20,2:18,3:16,4:16,5:16})

# — Sheet 6: Forecasting
print(' Building Sheet 6: Forecasting...')
ws6=wb.create_sheet('Forecasting 2025-2030')
xl_title(ws6,'Unclaimed Data Dividend: Forecasting Model (2025-2030)',
    'Three scenarios from $140M base (2025) | Creator payout = $0 throughout')
xl_hdr(ws6,4,range(1,8),['Year','Conservative\n(20% CAGR)','Base Case\n(30% CAGR)',
    'Optimistic\n(40% CAGR)','Creator Payout','Gap (Base)','Cum Gap (Base)'])
xl_colw(ws6,{1:8,2:20,3:18,4:18,5:18,6:16,7:20})
cum_g=0
for i,yr in enumerate(proj_years):
    vals=[round(base_2025*(1+r)**i,1) for r in [0.20,0.30,0.40]]
    gap=vals[1]; cum_g+=gap
    xl_row(ws6,5+i,range(1,8),
        [yr]+[f'${v:.0f}M' for v in vals]+[f'$0',f'${gap:.0f}M',f'${cum_g:.0f}M'],
        bg=VLIGHT if i%2==0 else WHITE)
    ws6.cell(5+i,5).fill=xl_fill('FFE5E5')
    ws6.cell(5+i,5).font=Font(name='Calibri',bold=True,color=RED_HEX)
    ws6.cell(5+i,7).fill=xl_fill('FFF0F0')
    ws6.cell(5+i,7).font=Font(name='Calibri',bold=True,color=ACCENT_HEX)

# — Sheet 7: Platform Exploitation
print(' Building Sheet 7: Platform Exploitation...')
ws7=wb.create_sheet('Platform Exploitation Index')
xl_title(ws7,'Platform Exploitation Index – Full Analysis',
    'Revenue split comparison | Source: SEC Filings · Corporate Annual Reports')
xl_hdr(ws7,4,range(1,7),['Platform','Revenue Model','Platform Share %',
    'Creator Share %','Category','For Every $100'])
xl_colw(ws7,{1:22,2:22,3:18,4:16,5:22,6:30})
cat_fills_xl={'Equitable':'E8FDF2','Moderate':'FFF9C4',
    'Exploitative':'FFE0B2','Maximum Extraction':'FFE5E5'}
for i,(_,row) in enumerate(platforms.iterrows()):
    cat=row['Category']; bg=cat_fills_xl.get(cat,WHITE)
    xl_row(ws7,5+i,range(1,7),
        [row['Platform'],row['Revenue_Model'],row['Platform_Share'],
        row['Creator_Share'],cat,f'${row["Creator_Share"]} to creator'],bg=bg)
    ws7.cell(5+i,5).font=Font(name='Calibri',bold=True,
        color=RED_HEX if cat=='Maximum Extraction' else ('1a7a1a' if cat=='Equitable' else '000000'))
ws7.conditional_formatting.add('D5:D11',
    DataBarRule(start_type='min',end_type='max',start_value=0,end_value=70,color='2ECC71'))

# — Sheet 8: Gini Data
print(' Building Sheet 8: Gini & Lorenz...')
ws8=wb.create_sheet('Gini & Lorenz Data')
xl_title(ws8,'Gini Coefficient & Lorenz Curve Data',
    'Creator income inequality | Source: Praneeth & Tamizharasi (2025)')
xl_hdr(ws8,4,range(1,6),['Income Bracket','Creators','Midpoint ($)','Total Income','Income Share %'])
xl_colw(ws8,{1:16,2:14,3:16,4:20,5:16})
for i,(_,row) in enumerate(income_brackets.iterrows()):
    bg=VLIGHT if i%2==0 else WHITE
    xl_row(ws8,5+i,range(1,6),
        [row['Bracket'],row['Creators'],row['Midpoint_USD'],
        row['Total_Income'],f'{row["Income_Share"]:.2f}%'],bg=bg)
ws8.cell(12,1).value=f'GINI COEFFICIENT = {creator_gini:.4f}'
ws8.cell(12,1).font=Font(name='Calibri',bold=True,size=13,color=ACCENT_HEX)
ws8.merge_cells('A12:E12')
ws8.cell(12,1).fill=xl_fill('FFE5E5')
ws8.cell(12,1).alignment=xl_center()
ws8.row_dimensions[12].height=28
xl_hdr(ws8,14,range(1,5),['Gini Comparison Entity','Gini Coefficient',
    'vs Creator Economy','Context'])
for i,(_,row) in enumerate(national_ginis.iterrows()):
    diff=creator_gini-row['Gini']
    bg='E8FDF2' if row['Gini']<0.5 else ('FFF0F0' if row['Gini']>=0.7 else VLIGHT)
    xl_row(ws8,15+i,range(1,4),
        [row['Entity'],f'{row["Gini"]:.3f}',
        f'+{diff:.3f}' if diff>0 else f'{diff:.3f}'],bg=bg)

# — Sheet 9: AI Deals
print(' Building Sheet 9: AI Deals...')
ws9=wb.create_sheet('AI Licensing Deals')
xl_title(ws9,'Confirmed AI Data Licensing Deals',
    'Source: Reddit S-1 · Reuters · Bloomberg · Digiday · Authors Alliance (2024)')
xl_hdr(ws9,4,range(1,7),['Platform','Industry','AI Buyer',
    'Deal Value','Creator Share','Year'])
xl_colw(ws9,{1:18,2:16,3:16,4:24,5:16,6:8})
deal_rows=[
    ('Reddit','Social Media','Google','$60M/year','$0',2024),
    ('Reddit','Social Media','OpenAI','$70M/year','$0',2024),
    ('Reddit','Social Media','Multiple','$140M (2025 total)','$0',2025),
    ('News Corp','News Media','OpenAI','$250M over 5 years','$0',2024),
    ('Wiley','Academic','Undisclosed','$44M total','$0',2024),
    ('Financial Times','News Media','OpenAI','$5-10M/year','$0',2024),

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('AP', 'News Media', 'OpenAI', 'Undisclosed', '$0', 2023),
]
for i, rd in enumerate(deal_rows):
    bg=VLIGHT if i%2==0 else WHITE; rn=5+i
    xl_row(ws9, rn, range(1,7), list(rd), bg=bg)
    ws9.cell(rn, 5).fill=xl_fill('FFE5E5')
    ws9.cell(rn, 5).font=Font(name='Calibri', bold=True, color=RED_HEX)
ws9.cell(13, 1).value=('KEY FINDING: Across ALL documented deals, creator payout = $0. '
    '100% of AI licensing revenue is captured by platform intermediaries.')
ws9.cell(13, 1).font=Font(name='Calibri', bold=True, color=ACCENT_HEX, italic=True)
ws9.merge_cells('A13:F13')
ws9.cell(13, 1).fill=xl_fill('FFE5E5')
ws9.cell(13, 1).alignment=Alignment(horizontal='left', vertical='center', wrap_text=True)
ws9.row_dimensions[13].height=30

xl_path = os.path.join(XLDIR, 'DataDividend_Analytics_Workbook.xlsx')
wb.save(xl_path)
print(f' Excel workbook saved → {xl_path}')
print(f' Sheets: {len(wb.sheetnames)}')

# =====
# MODULE 06 – ADVANCED & UNIQUE VISUALIZATIONS (A01-A10)
# =====

print('\n' + '='*70)
print('MODULE 06 – ADVANCED & UNIQUE VISUALIZATIONS (A01-A10)')
print('='*70)

# — A01: Sankey Money Flow
fig, ax = plt.subplots(figsize=(14, 8))
ax.set_xlim(0, 10); ax.set_ylim(-0.5, 10)
ax.axis('off'); fig.patch.set_facecolor('white')

def fbox(ax, x, y, w, h, color, text, sub='', tc='white', fs=11):
    box=FancyBboxPatch((x, y), w, h, boxstyle='round', pad=0.15,
        facecolor=color, edgecolor='white', lw=2, zorder=5)
    ax.add_patch(box)
    ax.text(x+w/2, y+h/2+(0.15 if sub else 0), text, ha='center', va='center',
        fontsize=fs, fontweight='bold', color=tc, zorder=6)
    if sub:
        ax.text(x+w/2, y+h/2-0.3, sub, ha='center', va='center',
            fontsize=9, color=tc, alpha=0.9, zorder=6)

ax.text(5, 9.5, 'The Data Dividend Flow: Where Does the Money Go?',
    ha='center', va='center', fontsize=15, fontweight='bold', color=DARK)
ax.text(5, 9.1, 'Tracing $130M of Reddit AI licensing revenue from AI company to creator',
    ha='center', va='center', fontsize=10, color=GRAY)
fbox(ax, 0.2, 5.5, 2.2, 2.5, NAVY, 'AI Companies', '$130M/year\npaid for content', fs=10)
ax.annotate('', xy=(3.1, 6.75), xytext=(2.4, 6.75),
    arrowprops=dict(arrowstyle='->', color=NAVY, lw=4))
ax.text(2.75, 7.1, '$130M', ha='center', fontsize=10, fontweight='bold', color=NAVY)
fbox(ax, 3.1, 4.5, 2.3, 4.5, ACCENT, 'Platform\n(Reddit)', 'CAPTURES\n$130M (100%)', fs=10)
ax.annotate('', xy=(7.0, 8.2), xytext=(5.4, 7.5),
    arrowprops=dict(arrowstyle='->', color=ORANGE, lw=4))
ax.text(6.2, 8.0, '$130M', ha='center', fontsize=10, fontweight='bold', color=ORANGE)
ax.annotate('', xy=(7.0, 6.2), xytext=(5.4, 6.5),
    arrowprops=dict(arrowstyle='->', color=PURPLE, lw=3))
ax.annotate('', xy=(7.0, 4.2), xytext=(5.4, 5.3),
    arrowprops=dict(arrowstyle='->', color=GREEN, lw=2, linestyle='dashed'))
fbox(ax, 7.0, 7.5, 2.6, 1.4, ORANGE, 'Shareholders\n& Investors', '$130M (100%)', fs=9)
fbox(ax, 7.0, 5.6, 2.6, 1.3, PURPLE, 'Operations', 'Server costs,\nStaff, Marketing', fs=9)
fbox(ax, 7.0, 3.5, 2.6, 1.8, GREEN, 'Content Creators', 'RECEIVES: $0.00\nfrom AI licensing', fs=10)
ax.text(6.3, 4.55, 'X', ha='center', va='center', fontsize=28, color=RED, fontweight='bold', zorder=10)
ax.text(6.3, 4.1, 'BLOCKED', ha='center', fontsize=8.5, color=RED, fontweight='bold', zorder=10)
for x, y, w, h, col, txt, txcol in [
    (0.2, 1.5, 2.2, 1.5, TEAL, 'Google paid\n$60M/year', 'white'),
    (2.6, 1.5, 2.2, 1.5, TEAL, 'OpenAI paid\n$70M/year', 'white'),
    (5.0, 1.5, 2.2, 1.5, GOLD, 'Total: $130M/year', DARK),
    (7.4, 1.5, 2.2, 1.5, GREEN, 'Creator gets:\n$0/year', 'white'),
]:
    fbox(ax, x, y, w, h, col, txt, tc=txcol, fs=9)
ax.text(5, 0.8, 'Source: Reddit S-1 IPO Prospectus (SEC.gov) · Reddit Q4 2025 Earnings',
    ha='center', fontsize=8.5, color=GRAY, style='italic')
fig.tight_layout(); savefig(fig, 'A01_Sankey_Money_Flow.png')

# — A02: Radar Chart
categories_r = ['Revenue\nSharing', 'Transparency', 'Creator\nControl',
    'Opt-Out\nRight', 'Audit\nMechanism', 'Legal\nClarity']
N_r = len(categories_r)
angles_r = np.linspace(0, 2*np.pi, N_r, endpoint=False).tolist()
angles_r += angles_r[:1]
plat_radar = {
    'YouTube\n(Ad Model)': [8.5, 7.0, 6.0, 5.0, 6.5, 7.0],
    'Spotify\n(Royalties)': [9.0, 8.0, 7.0, 7.5, 8.5, 8.0],
    'Reddit\n(AI Licensing)': [0.5, 2.0, 1.0, 2.0, 1.5, 2.5],
    'DARA\n(Proposed)': [7.5, 9.5, 8.5, 9.0, 9.5, 8.5],
}
fig, ax = plt.subplots(figsize=(10, 10), subplot_kw=dict(polar=True))
for (name, vals), col in zip(plat_radar.items(), [NAVY, GREEN, ACCENT, TEAL]):
    v = vals + vals[:1]
    ax.plot(angles_r, v, color=col, lw=2.5, label=name)
    ax.fill(angles_r, v, color=col, alpha=0.12)
ax.set_xticks(angles_r[:1]); ax.set_xticklabels(categories_r, fontsize=10.5, fontweight='bold')
ax.set_ylim(0, 10); ax.set_yticks([2.4, 6.8, 10])

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ax.set_title('Platform Fairness Radar: Multi-Dimension Equity Assessment\n'
            'Score 0-10 | Author Assessment Based on SEC Filings & Policies',pad=20)
ax.legend(loc='upper right',bbox_to_anchor=(1.35,1.15),frameon=False,fontsize=10)
ax.set_facecolor('white'); ax.grid(color=GRAY,linestyle='--',alpha=0.4)
savefig(fig,'A02_Radar_Platform_Fairness.png')

# — A03: Bubble Chart
bubble_data=[
    ('YouTube', 31.5, 55, 0, NAVY, '55% creator'),
    ('Spotify', 19.3, 68, 0, GREEN, '68% creator'),
    ('Steam', 8.5, 70, 0, TEAL, '70% creator'),
    ('Roblox', 3.6, 28, 0, GOLD, '28% creator'),
    ('Reddit(Ads)', 1.1, 5, 0, ORANGE, '5% creator'),
    ('Reddit(AI)', 0.14, 0, 130, ACCENT, '0% creator'),
    ('News Corp', 0.25, 0, 50, RED, '0% creator'),
]

fig,ax=plt.subplots(figsize=(13,8))
for plat,rev,cpct,deal,col,lbl in bubble_data:
    sz=max(deal,rev)*60+200
    ax.scatter([rev],[cpct],s=sz,c=col,alpha=0.75,edgecolors='white',lw=2.5,zorder=5)
    oy=3 if cpct>30 else -6
    ax.annotate(plat,(rev,cpct),xytext=(rev+0.3,cpct+oy),
                fontsize=9.5,fontweight='bold',color=col,
                arrowprops=dict(arrowstyle='-',color=col,lw=1))
ax.axhline(35,color=GRAY,ls='--',lw=1.5,alpha=0.5)
ax.axvline(5,color=GRAY,ls='--',lw=1.5,alpha=0.5)
ax.text(20,38,'EQUITABLE GIANTS',fontsize=9,color=GREEN,style='italic',alpha=0.7)
ax.text(0.1,2,'EXTRACTIVE / AI DEALS',fontsize=9,color=RED,style='italic',alpha=0.7)
ax.set_xlabel('Annual Platform Revenue ($B USD equivalent)')
ax.set_ylabel('Creator Revenue Share (%)')
ax.set_title('Bubble Chart: Platform Revenue × Creator Share × AI Deal Size\n'
            'Bubble size ∝ AI deal size | Source: SEC Filings · Annual Reports')
ax.set_facecolor(BG); ax.set_xlim(-1,35); ax.set_ylim(-5,80)
leg=[mpatches.Patch(color=c,label=n)
     for n,c in [(('YouTube',NAVY),('Spotify',GREEN),('Steam',TEAL),('Roblox',GOLD),
                  ('Reddit(Ads)',ORANGE),('Reddit(AI)',ACCENT),('News Corp',RED))]]
ax.legend(handles=leg,frameon=False,fontsize=9,loc='upper right')
fig.tight_layout(); savefig(fig,'A03_Bubble_Chart.png')

# — A04: Three Eras Timeline
fig,ax=plt.subplots(figsize=(16,7))
ax.set_xlim(1998,2031); ax.set_ylim(-3.5,5)
ax.axis('off'); fig.patch.set_facecolor('white')
ax.axhline(0,color=DARK,lw=3,zorder=3)
for x1,x2,col,lbl,tc in [
    (1998,2014,'#EBF5FB','ERA 1: ATTENTION ECONOMY\n(1998–2014)',NAVY),
    (2014,2022,'#FEF9E7','ERA 2: ENGAGEMENT ECONOMY\n(2014–2022)',GOLD),
    (2022,2030,'#FDEDEC','ERA 3: EXTRACTION ECONOMY\n(2022–Present)',ACCENT),
]:
    ax.axvspan(x1,x2,ymin=0.28,ymax=0.95,alpha=0.35,color=col,zorder=1)
    ax.text((x1+x2)/2,3.8,lbl,ha='center',fontsize=10.5,fontweight='bold',color=tc,zorder=5)
for yr,y,lbl,col in [
    (2000,-0.8,'Google\nAds',NAVY),(2005,-1.5,'YouTube\nlaunched',NAVY),
    (2008,-0.8,'Ad targeting\nmatures',GOLD),(2018,-1.5,'Data\nscandals',GOLD),
    (2022,-0.8,'ChatGPT\nlaunched',ACCENT),(2024,-1.5,'Reddit\n$203M deal',ACCENT),
    (2025,-0.8,'Reddit\n$2.2B rev',ACCENT),
]:
    ax.plot([yr,yr],[0,y],color=col,lw=1.5,zorder=4,ls='--',alpha=0.7)
    ax.scatter([yr],[0],color=col,s=80,zorder=6,edgecolor='white',lw=1.5)
    ax.text(yr,y-0.25,lbl,ha='center',fontsize=8,color=col,fontweight='bold')
for yr,y,lbl,col in [
    (2008,1.0,'YouTube\nPartner Prog',GREEN),(2024,1.8,'DARA\nProposed',TEAL),
    (2026,0.8,'EU AI Act\nEnforced',PURPLE),(2028,1.8,'DARA\nLaunch?',TEAL),
]:
    ax.plot([yr,yr],[0,y],color=col,lw=1.5,zorder=4,alpha=0.7)
    ax.scatter([yr],[0],color=col,s=80,zorder=6,edgecolor='white',lw=1.5)
    ax.text(yr,y+0.15,lbl,ha='center',fontsize=8,color=col,fontweight='bold')
for yr in range(2000,2031,2):
    ax.text(yr,-0.25,str(yr),ha='center',fontsize=8,color=GRAY)
ax.set_title('Three Eras of Internet Value: Attention → Engagement → Extraction (1998–2030)',
            fontsize=13,fontweight='bold',y=0.98)
savefig(fig,'A04_Three_Eras_Timeline.png')

# — A05: Waterfall
cats_wf=['AI Company\nPays Reddit','Platform\nOperations','Executive\nComp.',
        'Shareholder\nReturns','Platform\nReserves','Marketing\n& Legal','Creator\nPayout']
vals_wf=[130,-28,-15,-42,-25,-20,0]
cols_wf=[NAVY,ORANGE,ORANGE,ORANGE,ORANGE,ORANGE,RED]
running=0; btms=[]; bvals=[]
for v in vals_wf:
    if v>0: btms.append(0); bvals.append(v); running=v
    elif v<0: btms.append(running+v); bvals.append(-v); running+=v
    else: btms.append(0); bvals.append(0)
fig,ax=plt.subplots(figsize=(13,7))
bars_wf=ax.bar(range(len(cats_wf)),bvals,bottom=btms,color=cols_wf,
                edgecolor='white',lw=2,width=0.6,zorder=5)
for i,(bar,ov,bot) in enumerate(zip(bars_wf,vals_wf,btms)):
    lbl=f'f+${bar.get_height():.0f}M' if ov>0 else (f'-${bar.get_height():.0f}M' if ov<0 else '$0')
    ax.text(bar.get_x()+bar.get_width()/2,bot+bar.get_height()/2,
            lbl,ha='center',va='center',fontsize=10,fontweight='bold',color='white')
for i in range(len(cats_wf)-1):
    top=btms[i]+bvals[i]
    ax.plot([i+0.3,i+0.7],[top,top],color=GRAY,lw=1.5,ls='--',zorder=4,alpha=0.7)

```

```

ax.text(0,138,'$130M\n(100%)',ha='center',fontsize=9.5,color=NAVY,fontweight='bold')
ax.text(6,-8,'$0\n(0%)',ha='center',fontsize=12,color=RED,fontweight='bold')
ax.axhline(0,color=GRAY,lw=1,alpha=0.5)
ax.annotate('Creator payout: $0',xy=(6,0),xytext=(4.5,40),
            arrowprops=dict(arrowstyle='->',color=RED,lw=2),
            fontsize=11,color=RED,fontweight='bold',
            bbox=dict(boxstyle='round,pad=0.4',facecolor='#FFE5E5',edgecolor=RED))
ax.set_xticks(range(len(cats_wf))); ax.set_xticklabels(cats_wf,fontsize=10)
ax.set_ylabel('Amount (USD Millions)'); ax.set_ylim(-20,155); ax.set_facecolor(BG)
ax.set_title('Waterfall: How $130M of AI Licensing Revenue Disappears Before Reaching Creators\n'
            'Source: Reddit S-1 · Author Allocation Estimate')
fig.tight_layout(); savefig(fig,'A05_Waterfall_Revenue_Flow.png')

# — A06: Creator Income Pyramid
bkts=['$1M\n(0.33%)','$500K-$1M\n(1.1%)','$100K-$500K\n(5.5%)',
      '$50K-$100K\n(11%)','$10K-$50K\n(27.5%)','$0-$10K\n(55%)']
crts=[300,1000,5000,10000,25000,50000]
ishrs=[13.79,17.24,28.74,17.24,17.24,5.75]
pcols=['#8B0000',RED,ACCENT,ORANGE,GOLD,TEAL]
fig,(ax1,ax2)=plt.subplots(1,2,figsize=(15,8))
mx=max(crts)
for i,(bkt,cnt,col) in enumerate(zip(bkts,crts,pcols)):
    y=len(bkts)-1-i; bw=cnt/mx*8
    ax1.barh(y,bw,color=col,alpha=0.85,edgecolor='white',lw=1.5,height=0.7)
    ax1.text(bw+0.15,y,f'{cnt:,}',va='center',fontsize=10,fontweight='bold')
    ax1.text(-0.1,y,bkt,va='center',ha='right',fontsize=9.5)
ax1.set_xlim(-3.5,10); ax1.set_ylim(-0.7,len(bkts)-0.3); ax1.axis('off')
ax1.set_title('Creator Population\n(Number of Creators per Income Bracket)',pad=10)
ms=max(ishrs)
for i,(bkt,shr,col) in enumerate(zip(bkts,ishrs,pcols)):
    y=len(bkts)-1-i; bw=shr/ms*8
    ax2.barh(y,bw,color=col,alpha=0.85,edgecolor='white',lw=1.5,height=0.7)
    ax2.text(bw+0.15,y,f'{shr:.1f}%',va='center',fontsize=10,fontweight='bold')
ax2.set_xlim(-0.5,10); ax2.set_ylim(-0.7,len(bkts)-0.3); ax2.axis('off')
ax2.set_title('Income Share\n(% of Total Creator Income)',pad=10)
ax2.text(4,-0.5,f'Gini = {creator_gini:.3f} (higher than any nation)',
        ha='center',fontsize=10,color=ACCENT,fontweight='bold',style='italic')
fig.suptitle('Creator Income Pyramid: The Inequality Behind the "Creator Economy"\n'
            'Source: Praneeth & Tamizharasi (2025) · influencers.club',
            fontsize=13,fontweight='bold',y=1.01)
savefig(fig,'A06_Creator_Income_Pyramid.png')

# — A07: Correlation Matrix
enc_a={'Not at all aware':1,'Vaguely aware':2,'Somewhat aware':3,'Highly aware':4}
enc_b={'Significantly decreases':1,'Somewhat decreases':2,'No change':3,'Indifferent':4,'Unsure':3}
enc_r={'No platforms provide value':1,'No impractical':2,'Yes opt-out':3,
      'Yes conditional transparent':4,'Yes fully support':5}
enc_p={'No one':1,'Users/Creators':2,'Government':3,'AI Companies':4,'Platforms':5}
df_enc=pd.DataFrame({
    'Awareness': df_full['Awareness'].map(enc_a),
    'Behavior': df_full['Behavior'].map(enc_b),
    'Fair Score': df_full['Score'],
    'Roy Support': df_full['Royalty'].map(enc_r),
    'Responsible': df_full['Responsible'].map(enc_p),
})
corr=df_enc.corr()
lbls=['Awareness\nLevel','Behavioral\nIntent','Fairness\nScore',
      'Royalty\nSupport','Responsible\nParty']
fig,ax=plt.subplots(figsize=(9,8))
im=ax.imshow(corr.values,cmap='RdYlGn',aspect='auto',vmin=-1,vmax=1)
plt.colorbar(im,ax=ax,label='Pearson r',shrink=0.85)
ax.set_xticks(range(5)); ax.set_xticklabels(lbls,fontsize=10)
ax.set_yticks(range(5)); ax.set_yticklabels(lbls,fontsize=10)
for i in range(5):
    for j in range(5):
        v=corr.values[i,j]
        stars='***' if abs(v)>0.5 else ('**' if abs(v)>0.3 else ('*' if abs(v)>0.15 else ''))
        ax.text(j,i,f'{v:.2f}{stars}',ha='center',va='center',fontsize=10.5,
                fontweight='bold',color='white' if abs(v)>0.6 else 'black')
ax.set_title('Survey Variable Correlation Matrix\n'
            '* p<0.15 ** p<0.10 *** p<0.05 | Source: Primary Survey n=100')
ax.set_facecolor(BG)
savefig(fig,'A07_Correlation_Matrix.png')

# — A08: Diverging Bar
resp_scores=(df_full.groupby('Responsible')['Score']
            .agg(['mean','std','count']).reset_index()
            .sort_values('mean'))
overall_m=df_full['Score'].mean()
resp_scores['deviation']=resp_scores['mean']-overall_m
bar_c2=[GREEN if d>0 else ACCENT for d in resp_scores['deviation']]
fig,ax=plt.subplots(figsize=(11,6))
bars_d=ax.barh(resp_scores['Responsible'],resp_scores['deviation'],
                color=bar_c2,edgecolor='white',lw=1.5,height=0.55,zorder=5)
for bar,mn,n in zip(bars_d,resp_scores['mean'],resp_scores['count']):
    xp=bar.get_width()
    lx=xp+0.03 if xp>0 else xp-0.03
    ha='left' if xp>0 else 'right'
    ax.text(lx,bar.get_y()+bar.get_height()/2,
            f'Mean={mn:.2f} (n={n})',va='center',ha=ha,fontsize=9.5,fontweight='bold')
ax.axvline(0,color=DARK,lw=2,zorder=6)
ax.text(0,5.25,f'Overall mean = {overall_m:.2f}',ha='center',fontsize=9,color=DARK,fontweight='bold')
ax.set_xlabel(f'Deviation from Overall Mean ({overall_m:.2f})')

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ax.set_title('Diverging Bar: Fairness Score Deviation by Responsible Party Group(n
    'Source: Primary Survey n=100')
ax.set_facecolor(BG); ax.set_xlim(-1.5,1.5)
leg2=[mpatches.Patch(color=GREEN, label='Above-average fairness perception'),
      mpatches.Patch(color=ACCENT, label='Below-average fairness perception')]
ax.legend(handles=leg2, frameon=False, fontsize=10)
fig.tight_layout(); savefig(fig, 'A08_Diverging_Fairness_Responsible.png')

# — A09: DARA Implementation Roadmap
phases=[
    ('Phase 1: Policy Advocacy', 2025, 1.0, 'Active', TEAL),
    ('Phase 2: Regulatory Framework', 2025, 1.5, 'In Progress', NAVY),
    ('Phase 3: Pilot Programme', 2026, 1.0, 'Planned', GOLD),
    ('Phase 4: Platform Onboarding', 2026, 1.5, 'Planned', ORANGE),
    ('Phase 5: Creator Registration', 2027, 0.5, 'Planned', PURPLE),
    ('Phase 6: First Distributions', 2027, 0.5, 'Planned', GREEN),
    ('Phase 7: Full Scale Launch', 2027, 3.0, 'Target', ACCENT),
    ('Phase 8: Annual Audits', 2028, 2.0, 'Ongoing', GRAY),
]
milestones_r=[(2025, 'DARA\nProposed'), (2026, 'EU AI Act'), (2027, 'First\nRoyalties'), (2030, 'Full\nScale')]
fig, ax = plt.subplots(figsize=(14, 8))
for i, (ph, st, dur, status, col) in enumerate(phases):
    y = len(phases) - 1 - i
    alpha = 0.9 if status in ['Active', 'In Progress'] else 0.65
    ax.barh(y, dur, left=st, color=col, alpha=alpha, edgecolor='white', lw=2, height=0.65, zorder=5)
    ax.text(st + dur/2, y, f'{ph}\n[{status}]', ha='center', va='center',
            fontsize=8.5, fontweight='bold', color='white', zorder=6)
for yr, lbl in milestones_r:
    ax.axvline(yr, color=DARK, lw=2, ls='--', alpha=0.6, zorder=4)
    ax.text(yr, len(phases) - 0.2, lbl, ha='center', fontsize=9, color=DARK, fontweight='bold',
            bbox=dict(boxstyle='round', pad=0.3, facecolor='white', edgecolor=DARK))
ax.set_xlim(2024.5, 2031)
ax.set_yticks(range(len(phases)))
ax.set_yticklabels([p[0] for p in reversed(phases)], fontsize=9.5)
ax.set_xlabel('Year')
ax.set_xticks(np.arange(2025, 2031))
ax.set_title('DARA Implementation Roadmap: Policy Proposal to Full Deployment\n'
            'Digital Authorship Rights Administration | Author Proposal')
ax.set_facecolor(BG)
leg3=[mpatches.Patch(color=TEAL, alpha=0.9, label='Active'),
      mpatches.Patch(color=NAVY, alpha=0.9, label='In Progress'),
      mpatches.Patch(color=GOLD, alpha=0.65, label='Planned'),
      mpatches.Patch(color=GREEN, alpha=0.65, label='Target'),
      mpatches.Patch(color=GRAY, alpha=0.65, label='Ongoing')]
ax.legend(handles=leg3, frameon=False, fontsize=9, loc='lower right')
fig.tight_layout(); savefig(fig, 'A09_DARA_Roadmap.png')

# — A10: Summary Infographic Dashboard
fig = plt.figure(figsize=(18, 12))
fig.patch.set_facecolor(DARK)
ax_t = fig.add_axes([0, 0.93, 1, 0.07])
ax_t.set_facecolor(NAVY); ax_t.axis('off')
ax_t.text(0.5, 0.65, 'THE DATA DIVIDEND', ha='center', va='center',
          fontsize=22, fontweight='bold', color='white', transform=ax_t.transAxes)
ax_t.text(0.5, 0.2,
          'Who Owns the Internet You Built? | IIM Rohtak IPM06164 | Aditya Pratap Singh Chandel',
          ha='center', va='center', fontsize=11, color='#BDC3C7', transform=ax_t.transAxes)
def kpi(fig, pos, val, lbl, sub, col):
    ax = fig.add_axes(pos); ax.set_facecolor(col); ax.axis('off')
    ax.text(0.5, 0.65, val, ha='center', va='center', fontsize=22, fontweight='bold',
            color='white', transform=ax.transAxes)
    ax.text(0.5, 0.30, lbl, ha='center', va='center', fontsize=10.5, fontweight='bold',
            color='white', transform=ax.transAxes)
    ax.text(0.5, 0.08, sub, ha='center', va='center', fontsize=8.5, color='#BDC3C7',
            transform=ax.transAxes)
kpi(fig, [0.01, 0.76, 0.18, 0.15], '$203M', 'Reddit AI Deal', '2024 S-1 Filing', ACCENT)
kpi(fig, [0.21, 0.76, 0.18, 0.15], '$0', 'Creator Payout', 'From AI licensing', '#8B0000')
kpi(fig, [0.41, 0.76, 0.18, 0.15], f'{creator_gini:.3f}', 'Creator Gini', 'Higher than any nation', PURPLE)
kpi(fig, [0.61, 0.76, 0.18, 0.15], f'{sup_pct:.0f}%', 'Support DARA', 'Primary survey n=30', GREEN)
kpi(fig, [0.81, 0.76, 0.18, 0.15], f'${dara_pool:.0f}M', 'DARA Pool', 'Annual at 5% fee', TEAL)
ax_p = fig.add_axes([0.01, 0.48, 0.28, 0.26]); ax_p.set_facecolor('#1a2a4a'); ax_p.axis('off')
ax_p.pie([100, 0.001], colors=[ACCENT, '#2c2c2c'], startangle=90,
         wedgeprops=dict(edgecolor='white', lw=2))
ax_p.text(0, 0, 'Reddit\nAI Deal', ha='center', va='center', fontsize=11,
          fontweight='bold', color='white')
ax_p.text(0, -1.35, 'Revenue Split: 100% Platform / 0% Creator',
          ha='center', fontsize=9, color='#BDC3C7')
ax_p.set_title('Revenue Split', color='white', fontsize=11, fontweight='bold', pad=5)
ax_b = fig.add_axes([0.36, 0.48, 0.28, 0.26]); ax_b.set_facecolor('#1a2a4a')
btm = 0
for lbl, pct, col in [('Full Support', 10.0, GREEN), ('Cond(Audit)', 43.3, '#27ae60'),
                     ('Cond(Opt-Out)', 30.0, '#f39c12'), ('Not Practical', 6.7, ACCENT),
                     ('Platforms OK', 10.0, '#c0392b')]:
    ax_b.bar([0], pct, bottom=btm, color=col, alpha=0.8, edgecolor='white', lw=1)
    if pct > 8:
        ax_b.text(0, btm + pct/2, f'{pct:.0f}%', ha='center', va='center',
                  fontsize=9, fontweight='bold', color='white')
    btm += pct
ax_b.set_xlim(-0.5, 0.5); ax_b.set_ylim(0, 105); ax_b.axis('off')
ax_b.text(0, 105, 'Royalty Support', ha='center', fontsize=10, fontweight='bold', color='white')
ax_b.text(0, -8, f'{sup_pct:.0f}% support DARA in some form',
          ha='center', fontsize=9, color='#BDC3C7')
ax_g = fig.add_axes([0.71, 0.48, 0.28, 0.26]); ax_g.set_facecolor('#1a2a4a')
gn = ['Sweden', 'India', 'USA', 'Brazil', 'Creator\nEconomy', 'Reddit\nAI']

```

```
gv=[0.28,0.35,0.41,0.53,creator_gini,1.00]
gc=[GREEN,TEAL,TEAL,GOLD,ACCENT,'#8B0000']
bg3=ax_g.barh(gn,gv,color=gc,edgecolor='white',lw=1,height=0.6)
for bar,v in zip(bg3,gv):
    ax_g.text(v+0.01,bar.get_y()+bar.get_height()/2,f'{v:.2f}',
              va='center',fontSize=9,fontWeight='bold',
              color='white' if v>0.7 else '#BDC3C7')
ax_g.set_xlim(0,1.15); ax_g.set_facecolor('#1a2a4a')
ax_g.tick_params(colors='#BDC3C7',labelsize=9)
ax_g.spines['bottom'].set_color('#BDC3C7')
for sp in ['top','right','left']:
    ax_g.spines[sp].set_visible(False)
ax_g.set_title('Gini Comparison',color='white',fontSize=10,fontWeight='bold',pad=5)
ax_bot=fig.add_axes([0,0.02,1,0.07])
ax_bot.set_facecolor(ACCENT); ax_bot.axis('off')
ax_bot.text(0.5,0.6,
            'KEY FINDING: Platforms earned $130M+ in AI licensing from creators\' content. '
            'Creators received $0.',
            ha='center',va='center',fontSize=13,fontWeight='bold',color='white',
            transform=ax_bot.transAxes)
ax_bot.text(0.5,0.18,
            'Source: Reddit S-1 IPO Prospectus (SEC.gov) · Spotify 20-F · Primary Survey n=30 · Author Analysis',
            ha='center',va='center',fontSize=9,color='#FFD5DC',transform=ax_bot.transAxes)
savefig(fig,'A10_Summary_Infographic.png')

# =====
# FINAL SUMMARY
# =====

print('\n' + '*70)
print('MASTER ANALYTICS COMPLETE')
print('*70)
print(f"""
Output directory: {ROOT}
├── figures/          26 charts (F01-F06, S01-S10, A01-A10)
│   ├── Financial:    F01 Platform Exploitation Index
│   │                 F02 Reddit Revenue Divergence 2020-2025
│   │                 F03 Spotify Revenue vs Royalties
│   │                 F04 AI Deal Landscape (Lollipop)
│   │                 F05 Value Split Comparison
│   │                 F06 Revenue Era Comparison
│   ├── Statistical:  S01 Regression Comparison
│   │                 S02 Lorenz Curve
│   │                 S03 Gini Benchmark
│   │                 S04 Forecasting Fan Chart
│   │                 S05 Survey 4-Panel Dashboard
│   │                 S06 Fairness Distribution + Violin
│   │                 S07 Awareness vs Behavior Heatmap
│   │                 S08 Awareness vs Responsible Heatmap
│   │                 S09 Box Plot Fairness by Awareness
│   │                 S10 Royalty Support Stacked by Awareness
│   ├── Advanced:    A01 Sankey Money Flow
│   │                 A02 Radar Chart Platform Fairness
│   │                 A03 Bubble Chart
│   │                 A04 Three Eras Timeline
│   │                 A05 Waterfall Revenue Flow
│   │                 A06 Creator Income Pyramid
│   │                 A07 Correlation Matrix
│   │                 A08 Diverging Bar Fairness
│   │                 A09 DARA Implementation Roadmap
│   └── A10 Summary Infographic Dashboard
├── excel/
│   └── DataDividend_Analytics_Workbook.xlsx (9 sheets)
├── data/             6 CSV data files
└── master_statistics.csv

Key Findings:
Reddit AI Revenue (2025)      : $140M | Creator Payout: $0
Spotify Royalty Correlation   : r = {spot_r:.3f} | Reddit AI: r = 0.000
Creator Economy Gini          : {creator_gini:.3f} | Reddit AI Gini: 1.000
Survey Support for DARA       : {sup_pct:.0f}% (actual n=30)
Unclaimed Dividend 2025-2030 : ${cum_base/1000:.2f}B (base case)
DARA Annual Pool              : ${dara_pool:.0f}M at 5% fee
""")
```

	F02 Reddit Revenue Divergence 2020-2025
	F03 Spotify Revenue vs Royalties
	F04 AI Deal Landscape (Lollipop)
	F05 Value Split Comparison
	F06 Revenue Era Comparison
Statistical:	S01 Regression Comparison
	S02 Lorenz Curve
	S03 Gini Benchmark
	S04 Forecasting Fan Chart
	S05 Survey 4-Panel Dashboard
	S06 Fairness Distribution + Violin
	S07 Awareness vs Behavior Heatmap
	S08 Awareness vs Responsible Heatmap
	S09 Box Plot Fairness by Awareness
	S10 Royalty Support Stacked by Awareness
Advanced:	A01 Sankey Money Flow
	A02 Radar Chart Platform Fairness
	A03 Bubble Chart
	A04 Three Eras Timeline
	A05 Waterfall Revenue Flow
	A06 Creator Income Pyramid
	A07 Correlation Matrix
	A08 Diverging Bar Fairness
	A09 DARA Implementation Roadmap
	A10 Summary Infographic Dashboard
excel/	
	DataDividend_Analytics_Workbook.xlsx (9 sheets)
data/	6 CSV data files
master_statistics.csv	
Key Findings:	
Reddit AI Revenue (2025)	: \$140M Creator Payout: \$0
Spotify Royalty Correlation	: r = 0.987 Reddit AI: r = 0.000
Creator Economy Gini	: 0.738 Reddit AI Gini: 1.000
Survey Support for DARA	: 83% (actual n=30)
Unclaimed Dividend 2025-2030	: \$1.79B (base case)
DARA Annual Pool	: \$155M at 5% fee